

**Web-Based Event Management System with Machine learning based
Facial Attendance for Student Organization of the National
Aviation Academy of the Philippines**

A Capstone Project

Presented to

Institute of Computer Studies Department

National Aviation Academy of the Philippines

(Formerly Philippine State College of the Philippines)

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CHAPTER I

INTRODUCTION

Attendance monitoring plays an important role in measuring participation and engagement in school and organizational activities. Many institutions still rely on manual attendance methods such as sign-in sheets and simple digital lists, which are often slow, prone to manipulation, and susceptible to human error. Studies have shown that manual attendance systems are inefficient for large events and may result in inaccurate or unreliable records due to human error and lack of real-time verification [1], [4]. In higher education institutions, particularly in student organization activities, accurate attendance monitoring is essential for documenting participation and evaluating student involvement. However, many organizations continue to use inconsistent attendance procedures that do not verify the identity of participants, leading to administrative inefficiencies and unreliable documentation.

Recent studies highlight the effectiveness of facial recognition technology in improving attendance accuracy and reducing fraudulent practices such as proxy attendance [7], [8], [10]. Facial recognition systems utilize machine learning and computer vision algorithms to automatically identify individuals and record their attendance in real time, significantly reducing manual intervention and improving verification accuracy [7], [8]. Despite these advantages, many existing attendance systems are primarily designed for classroom or workplace environments and are rarely integrated with centralized web-based event management platforms that can support large-scale organizational activities [5], [9]. Additionally, several systems rely on a single authentication method and lack hybrid verification approaches such as combining facial recognition with QR code fallback to ensure system reliability during technical limitations. Furthermore, limited studies evaluate attendance systems using recognized software quality standards such as ISO/IEC

25010, leaving gaps in understanding their usability, security, and overall system performance [3].

At the National Aviation Academy of the Philippines (NAAP), student organization events commonly rely on manual attendance sheets to record participant attendance. This approach may lead to issues such as proxy attendance, registration delays, incomplete monitoring of student participation, and misplaced attendance records, resulting in unreliable documentation and increased workload for student officers and faculty members. These challenges highlight the need for a more efficient attendance management approach that can ensure accurate verification and streamlined event monitoring. Therefore, the development of a web-based event management system integrated with machine learning-based facial recognition technology can help automate attendance verification, improve monitoring accuracy, and address the limitations of traditional attendance methods used in NAAP student organization events.

Project Context

At the National Aviation Academy of the Philippines (NAAP), student organizations regularly conduct academic, leadership, and co-curricular events where attendance monitoring is required for documentation and evaluation purposes. Currently, these organizations rely on manual sign-in sheets and separate spreadsheet records to monitor participation. This fragmented approach often leads to duplicated, misplaced, or incomplete data, particularly during large-scale events. Studies indicate that the absence of a centralized event management system can slow down data processing and make reporting more difficult, ultimately affecting the efficiency of organizational operations [5], [9].

The existing manual process also lacks reliable identity verification mechanisms. Attendance records may become inconsistent due to human error or

fraudulent practices such as proxy attendance, where students sign in on behalf of absent participants. Research shows that biometric technologies, particularly facial recognition systems, can significantly improve attendance accuracy by automating identity verification and reducing the need for manual intervention [7], [8], [10]. Implementing such a system within NAAP would help ensure that attendance records are both centralized and authenticated, improving the reliability of documentation for student organization events.

Purpose and Description of the Project

The system incorporates a machine learning–based facial recognition module to automate attendance verification. Specifically, the facial recognition component utilizes the InsightFace deep learning framework with cosine similarity for facial feature matching, which has been shown to provide high accuracy in biometric identification systems [10]. The model works by capturing facial images through a camera, extracting unique facial embeddings using a convolutional neural network (CNN), and comparing these embeddings with registered student records stored in the database. Cosine similarity is then used to determine whether the captured face matches the stored facial template of a registered participant. This approach allows the system to perform real-time identity verification while minimizing manual intervention and reducing the possibility of proxy attendance during events.

To further improve verification reliability, the system also incorporates liveness detection, which ensures that the captured facial image originates from a real person rather than a photograph or digital replica. In situations where facial recognition may fail due to environmental factors such as poor lighting or camera limitations, the system provides a QR code fallback mechanism to ensure that attendance can still be recorded efficiently. By integrating machine learning–based facial recognition with alternative verification methods, the system aims to provide

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a reliable and secure attendance monitoring solution for student organization events at NAAP.

Contextual Diagram

This study adopts the Input–Process–Output (IPO) model as its conceptual framework. The IPO model illustrates how the required inputs are collected and transformed through system development processes. It also incorporates statistical treatment and data analysis to support the evaluation of the study. The framework demonstrates how these processes generate the intended system output for the proposed web-based student consultation and faculty appointment management system.

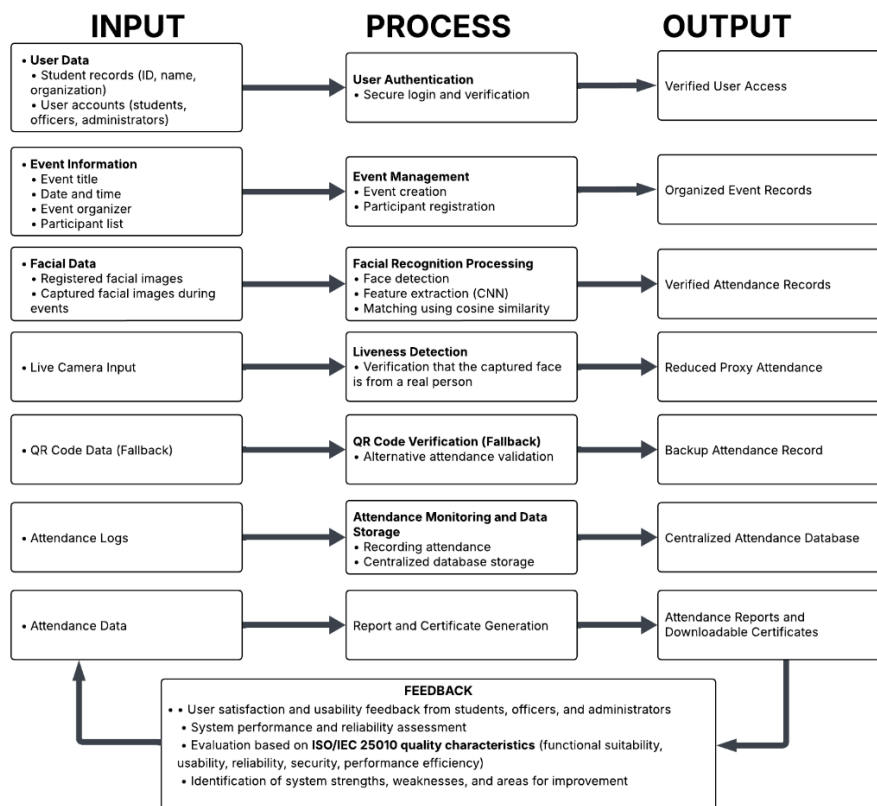


Figure 1: Input Process Output model

Figure 1 explains how the system works. The input includes student records, user accounts, event details, facial images, and attendance data needed by the system. The process begins with a secure way of logging the users in, creating events assigned to the user, and then face recognition for those events of this student's attendance list. After that, the system stores these data, analyzes attendance, generates reports, creates certificates, and enables qualified participants to access the certificate for download. Thus, the output will constitute the following elements: verified users, event records will be kept and organized; the attendance data will be accurate, reliable reports, downloadable certificates. In the performance, usability, and reliability testing of the system, user feedback may be regarded as useful.

Project Objectives

The general objective of this study is to evaluate the effectiveness of a web-based event management system in improving attendance monitoring, participation verification, and the efficiency of event documentation, including automated certificate generation, during student organization events at the National Aviation Academy of the Philippines (NAAP).

Specific Objectives

1. To evaluate the effectiveness of the hybrid attendance verification approach using facial recognition with QR code fallback in improving the accuracy and reliability of attendance tracking.
2. To determine the extent to which liveness detection helps prevent proxy attendance and fraudulent check-ins during attendance verification.

3. To assess how centralized attendance monitoring improves real-time tracking and management of event participants, including automated certificate generation for verified attendees.
4. To examine users' perceptions of the usability and efficiency of the attendance management system.
5. To evaluate the system based on the software quality characteristics defined in ISO/IEC 25010, particularly in terms of functional suitability, usability, reliability, security, and performance efficiency.
6. To gather and evaluate feedback from students, organization officers, and administrators to pinpoint the system's strengths, weaknesses, and areas for potential enhancement.

Scope and Limitations

This study focuses on the design, development, and evaluation of a web-based event management and attendance system with machine learning-based facial recognition for student organizations at the National Aviation Academy of the Philippines (NAAP). The system supports event management functions such as user authentication, event creation, participant registration, and attendance recording using facial recognition technology, with a QR code option provided as a backup verification method. The platform also includes centralized data storage, automated attendance reports, and certificate generation for verified participants. The system is intended for use by students, organization officers, and administrators who manage and participate in student organization events. System development and evaluation are conducted during selected NAAP student organization activities. The study follows a developmental research approach and evaluates the system based on the ISO/IEC 25010 software quality model,

focusing on functional suitability, usability, reliability, security, and performance efficiency [13], [17].

Despite its intended benefits, the study has several limitations. First, the system is developed and tested only within the National Aviation Academy of the Philippines, which may limit the generalizability of the results to other institutions with different operational environments. Second, the machine learning component of the system is limited to facial recognition and does not include other biometric or artificial intelligence technologies. Third, system testing and user feedback are conducted only during the duration of the study, which may not fully represent the system's long-term performance or scalability. The QR code attendance feature serves only as a fallback mechanism and is not evaluated as the primary attendance method.

Chapter II

Review of Related Literature

This chapter presents a review of studies and literature related to the development of a web-based event management and attendance system with machine learning–based facial recognition. The reviewed studies are grouped into thematic areas directly aligned with the objectives and conceptual framework of the present study. These themes include web-based event management systems, facial recognition–based attendance systems, machine learning applications in attendance monitoring, data integrity and security in academic systems, and software quality and usability standards.

Web-Based Event Management System

Web-based event management systems (EMS) are essential for centralizing event creation, participant registration, attendance tracking, and reporting in educational institutions. Studies show that these systems streamline event processes and reduce reliance on manual procedures. For instance, the UTHM EMS supports user registration, event creation, online payments, and QR code–based attendance monitoring [11].

Similarly, the NCDC SITAware system operates with low internet bandwidth, generating real-time notifications, reports, and dashboards for event accountability and postevent analysis [12]. Functional and user acceptance testing indicate that such systems improve efficiency, accessibility, and reliability while minimizing manual errors [11]. Similarly, broader literature reviews indicate that online event management platforms facilitate attendee coordination, schedule management, and interactive features that enhance participant satisfaction and event outcomes [13]. These studies collectively show that web-based systems improve operational effectiveness by automating routine tasks, maintaining centralized records, and

providing easy access to information for stakeholders. However, limitations remain in traditional web-based approaches. While QR code-based attendance tracking and manual verification improve speed, they are still vulnerable to issues such as proxy attendance, QR code sharing, and human errors during validation [11].

Building on these findings, integrating machine learning-based facial recognition can further improve attendance accuracy, prevent fraud, and enhance administrative efficiency. For the National Aviation Academy of the Philippines, a system combining centralized event management with intelligent facial recognition and automated reporting can resolve issues of proxy attendance, fragmented records, and inefficient manual workflows, improving both organizational performance and participant accountability.

Facial Recognition Attendance

In the era of digital transformation, traditional manual attendance systems in educational institutions are increasingly inefficient, time-consuming, and prone to fraudulent practices such as proxy attendance. Studies have highlighted that paper-based attendance often suffers from errors and intentional falsification, compromising the integrity of student records [16]–[18]. To address these challenges, facial recognition technology has emerged as a reliable biometric method for automating attendance verification.

Automated facial recognition systems use computer vision and machine learning algorithms, including CNNs, LBPH, and InsightFace, to identify and authenticate students in real time, reducing human intervention and improving accuracy, often exceeding 98% in controlled experiments [16], [17], [20]. These systems capture live images, extract unique facial features, and match them against pre-registered databases using similarity metrics like cosine similarity [20], ensuring attendance is recorded only when students are physically present, thus

minimizing fraudulent behaviors [16], [20]. Practical implementations, such as the Web-Based Attendance System with Facial Recognition (WAS-FR), provide real-time monitoring, centralized records, automated reporting, and classroom location tracking to prevent fake attendance [16], [18], [19], [20].

The evolution of these systems underscores the role of advanced machine learning models, such as FaceNet and InsightFace, in enhancing accuracy and scalability under varying conditions [10]. Privacy and data security remain crucial, with recommendations including data anonymization, encrypted storage, liveness detection, and opt-in consent to protect students' biometric information [10]. Overall, facial recognition-based attendance systems offer a highly accurate, automated, and scalable solution that improves administrative efficiency, reduces manual workloads, prevents fraud, and promotes accountability, aligning with institutional goals for enhanced academic engagement and operational effectiveness [16]–[20].

Authentication and Data Integrity

Authentication and data integrity are essential for secure attendance management in web-based systems, as they ensure the accuracy and reliability of recorded attendance. Traditional methods, including paper registers, logbooks, or QR code-based systems, are prone to errors, missing entries, and fraudulent practices such as proxy attendance, highlighting the need for automated verification mechanisms [21], [25].

Biometric and web-based applications that integrate authentication sensors such as fingerprint or facial recognition combined with centralized databases have significantly improved attendance accuracy and reduced opportunities for falsification [21], [22]. Fingerprint, facial, and iris recognition are particularly effective, as biometric traits are unique, difficult to replicate, and not easily

forgotten, thereby strengthening data integrity compared to passwords or PINs [22], [24], [25].

Implementations of fingerprint-based attendance systems using microcontrollers and sensors demonstrate reliable real-time verification, accurate database updates, and secure log storage within web platforms [23], [25]. Studies also emphasize that multi-factor and multimodal biometric authentication enhances overall system security and prevents unauthorized access or tampering [24], [25]. Integrating these mechanisms ensures accurate, tamper-resistant attendance records for the National Aviation Academy of the Philippines, reducing proxy attendance and safeguarding sensitive student information.

Attendance Analytics and Reporting Systems

Attendance analytics and reporting systems convert raw attendance records into meaningful information for monitoring participation, compliance, and institutional performance. Studies show that automated reporting improves data accuracy and timeliness by reducing manual consolidation and human errors. Modern systems integrate analytics features to support decision-making and operational oversight [26].

For example, the Online Attendance Management System [26] provides real-time updates, centralized databases, and automated reporting, enabling administrators to monitor attendance patterns, identify absenteeism, and access historical records efficiently. Facial recognition systems [27] enhance this with dashboards that visualize trends, punctuality, and student-wise summaries, while RFID-based solutions [28] enable large-scale real-time record management for compliance, audits, and strategic planning.

Web-based platforms, such as the Automated School Attendance and Analytics System [29], automate data capture and provide user-friendly monitoring

tools. Cloud-based facial recognition systems [30] further improve scalability, detect fraudulent attendance, generate aggregated reports, and offer fast, accessible interfaces. Collectively, these systems demonstrate that integrating real-time tracking, trend analysis, automated reporting, and fraud detection enhances attendance management, supports timely decision-making, and improves institutional efficiency.

Software Quality and Usability Standards

Ensuring high software quality and usability is essential for the effectiveness and acceptance of academic systems. The ISO/IEC 25010 standard provides a comprehensive framework for evaluating software quality, covering functional suitability, performance efficiency, security, reliability, maintainability, and usability [31], [32], [35]. Adherence to this standard enables developers and institutions to systematically assess system performance, identify improvement areas, and align software with international benchmarks.

Usability is a key factor influencing user acceptance in academic environments. High usability promotes adoption and continued use, while structured tools like Brooke's System Usability Scale (SUS) and well-designed survey scales provide reliable ways to measure subjective user experience [33]. Proper application ensures measurement validity, whereas misuse can undermine research quality. Hybrid approaches combining ISO/IEC 25010 with user perception frameworks, such as SERVQUAL, capture both technical characteristics and user-centered dimensions, highlighting gaps between system performance and user expectations [34], [35].

Advanced frameworks, like QualCode, integrate ISO/IEC 25010 submetrics with machine learning to assess maintainability, evaluating modularity, reusability, testability, analyzability, and modifiability [35]. Overall, these studies demonstrate

that combining recognized software quality standards with rigorous usability evaluation ensures academic systems are technically robust, user-friendly, and capable of enhancing effectiveness, reliability, and acceptance among students and faculty [31]–[35].

Facial Recognition Under Disguise and Makeup Conditions

Facial recognition systems have achieved high accuracy in controlled environments; however, their performance significantly declines when faces are altered through disguise and makeup [37]. These alterations affect both the structure and texture of facial features, making identification more difficult in real-world scenarios [38]. Disguises such as glasses, hats, masks, and wigs can obscure important facial regions, reducing the visibility of key features needed for accurate recognition [40]. Similarly, makeup can modify skin tone and facial appearance, resulting in inconsistencies in feature extraction and matching [31].

Traditional facial recognition methods that rely on handcrafted features, such as Principal Component Analysis (PCA), Local Binary Patterns (LBP), Scale-Invariant Feature Transform (SIFT), and Speeded-Up Robust Features (SURF), are particularly sensitive to these variations [40]. These techniques depend on stable and predefined facial patterns, making them less effective when faces are partially occluded or cosmetically altered [40].

In contrast, deep learning approaches, particularly Convolutional Neural Networks (CNNs), have shown improved performance by learning more complex and invariant facial representations [39]. These models are capable of adapting to variations in facial appearance caused by disguise and makeup, improving recognition accuracy under challenging conditions [39]. Evaluation of these methods is commonly conducted using challenging datasets such as the Disguise Faces and Makeup Dataset (DMFD), which include variations in disguise and

cosmetic alterations [37]. Similarly, datasets such as the Disguised Faces in the Wild (DFW) provide real-world scenarios with pose, lighting, and disguise variations [38]. Findings from these studies consistently show that deep learning models outperform traditional handcrafted methods in handling disguise and makeup conditions [39]. However, extreme cases of disguise and heavy makeup still pose limitations to facial recognition systems, indicating the need for further research to improve robustness and reliability [36].

Synthesis

Web-based event management systems have significantly enhanced administrative processes in educational institutions by centralizing event creation, participant registration, attendance tracking, and reporting. These systems improve efficiency, reduce manual workload, and provide real-time monitoring through automated check-in and centralized dashboards. However, traditional attendance verification methods such as QR codes and manual validation remain vulnerable to proxy attendance, code sharing, and human error. These limitations highlight the need for more secure, accurate, and automated approaches to attendance management.

To address these challenges, facial recognition based attendance systems powered by machine learning algorithms such as Convolutional Neural Networks (CNNs), LBPH, InsightFace, and FaceNet offer a more reliable solution. These systems enable real-time identity verification with high accuracy, minimizing fraudulent practices and improving accountability. When integrated into web-based platforms, they provide centralized data storage, automated reporting, and analytics for better decision-making. Additionally, the implementation of security measures such as encrypted storage, liveness detection, and user consent ensures the protection of sensitive biometric data while maintaining system integrity. However, challenges remain in real-world scenarios, particularly when

facial features are altered through disguise or makeup, which can reduce recognition accuracy despite improvements in deep learning models.

Overall, the integration of web-based event management with machine learning–driven facial recognition, supported by strong authentication mechanisms, data integrity measures, and adherence to software quality standards such as ISO/IEC 25010, provides a comprehensive solution to attendance management issues. These technologies enable the creation of secure, tamper-resistant, and user-friendly systems that improve operational efficiency and institutional accountability. By addressing problems such as proxy attendance, fragmented records, administrative inefficiency, and environmental limitations, the proposed system offers a scalable and effective solution tailored to the needs of the National Aviation Academy of the Philippines.

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Research Gap	Related Literature (RRL Theme)	Findings from Literature	Identified Limitation	Implication for Current Study
Lack of integrated systems combining event management with biometric-based attendance verification .	Theme 1: Web-Based Event Management System	Studies show that web-based EMS platforms automate event registration, scheduling, and attendance tracking using QR codes and centralized databases. [11]-[15]	Focus is limited to basic automation; lacks biometric-based verification and intelligent validation mechanisms, making systems vulnerable to proxy attendance and human error.	Development of a more advanced EMS integrating facial recognition for secure, automated, and fraud-resistant attendance monitoring.
Insufficient implementation of facial recognition in real-	Theme 2: Facial Recognition Attendance	Research demonstrates that facial recognition systems using machine	Performance is mostly evaluated in controlled environments with limited	Implementation of a real-world applicable facial recognition system

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world, large-scale academic systems integrated with web platforms.		learning (e.g., CNN, LBPH, InsightFace) achieve high accuracy and automate attendance recording [16]–[20].	consideration of real-world conditions such as lighting, crowding, and device variability.	integrated into a web-based platform with improved robustness and scalability.
Lack of comprehensive, multi-layered security frameworks combining biometrics, encryption, and centralized data integrity controls.	Theme 3: Authentication and Data Integrity	Biometric systems such as fingerprint and facial recognition improve authentication and ensure accurate attendance records [21]–[25].	Most systems rely on single-modal biometrics and lack comprehensive security measures such as multi-factor authentication, encryption, and secure access control.	Integration of secure authentication mechanisms and data integrity controls within a centralized system to ensure reliable and tamper-resistant attendance data.

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Absence of intelligent, predictive analytics integrated with biometric attendance data.	Theme 4: Attendance Analytics and Reporting Systems	Modern systems provide real-time attendance monitoring, dashboards, and automated reporting to support administrative decision-making [26]–[30].	Analytics are mostly descriptive, with limited predictive capabilities and weak integration with biometric data for fraud detection and deeper insights.	Development of an analytics-enabled system that leverages accurate biometric data for intelligent reporting, trend analysis, and decision support.
Lack of integrated evaluation frameworks combining software quality standards and usability for biometric-	Theme 5: Software Quality and Usability Standards	ISO/IEC 25010 and usability tools such as SUS are widely used to evaluate system quality and user experience in academic	Evaluation is often fragmented, focusing separately on technical quality or usability, with limited application in biometric-	Application of integrated evaluation frameworks combining ISO/IEC 25010 and usability metrics to ensure system effectiveness, efficiency, and

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based systems.		systems [31]–[35].	based systems.	user acceptance.
Lack of robust facial recognition systems that handle disguise and makeup in real-world applications.	Theme 6: Facial Recognition Under Disguise and Makeup Conditions	Deep learning models improve facial recognition under disguise and makeup, outperforming traditional methods [36]–[40].	Accuracy decreases under extreme disguise and heavy makeup.	Develop a system with improved robustness for real-world facial variations.

Chapter III

Technical Background

This section presents the technologies and tools used in developing the system. It also includes diagrams that illustrate the system architecture, data flow, and overall design to support better understanding of the system implementation.

Research and Development Methodology

The development of the Web-Based Event Management System with Machine Learning–Based Facial Attendance for the National Aviation Academy of the Philippines (NAAP) adopted an iterative Agile Software Development Life Cycle (SDLC) approach, specifically using a hybrid Scrum-inspired framework. Agile methodology was chosen because it allows continuous improvement, flexible changes, and step-by-step delivery of system features. Agile is widely used in web-based systems because it supports user-focused design and adapts easily to changing requirements [13].

The system involves different users, including student organization officers, members, and administrators. These users have different needs that may change during development. Agile supports continuous feedback and collaboration, which helps improve the system and ensure it meets user requirements.

Figure 2 illustrates the Agile development workflow adopted in this study, demonstrating the iterative progression of system development phases supported by continuous stakeholder feedback and improvement cycles.

Software Development Life Cycle Model (SDLC)



Figure 2: Agile Software Development Cycle Workflow for the Development of Web-Based Event Management System with Machine learning based Facial Attendance for Student Organization of the National Aviation Academy of the Philippines

Development Phases

The development of the Web-Based Event Management System with Machine Learning–Based Facial Attendance follows an iterative Agile approach, where the system is developed through a series of short development cycles called sprints. Each sprint delivers a functional component of the system, allowing continuous testing, feedback, and improvement.

Methodology Justification

The system in this study follows the Agile Development Model, which uses repeated cycles to develop and improve the system. Each cycle includes planning, design, development, testing, and deployment. This approach allows continuous feedback and helps ensure the system meets user needs. Agile is commonly used in web-based systems because it supports flexibility and continuous improvement [13].

System Development Framework

The system follows a layered architecture with presentation, application, and data layers. It uses PHP for backend processing, SQL for database management, and HTML, CSS, and JavaScript for the user interface. Python is used for the facial recognition component to process and verify attendance. This framework helps the system operate in an organized, efficient, and reliable way.

Analysis of the Existing System

The current way student events are managed at the National Aviation Academy of the Philippines (NAAP) relies on old, slow methods. While these methods work for small tasks, they are not fast or safe enough for a modern school.

Current Evaluation Procedures

Attendance during NAAP student organization events is currently recorded using manual sign-in sheets or basic digital spreadsheets. Event evaluation and documentation are performed separately, often requiring manual data consolidation.

The evaluation of attendance data is also highly dependent on human intervention, which increases the likelihood of inconsistencies and inaccuracies. Organization officers must manually verify attendance lists, cross-check names,

and consolidate records from multiple sheets or files. In many cases, handwriting issues, incomplete entries, and duplicate records complicate the validation process. Furthermore, the absence of real-time verification allows practices such as proxy attendance, where students sign in on behalf of others, undermining the credibility of attendance records used for certificates and organizational assessments.

Additionally, the current procedures lack centralized data management and automated reporting tools. Attendance data, event details, and participant information are stored separately and are not integrated into a single system. As a result, generating attendance summaries, evaluation reports, and certificates requires additional manual effort and is often delayed. These limitations affect the overall efficiency of student organization operations and reduce the reliability of attendance data used for institutional documentation and decision-making.

Limitations and Problem Analysis

In this phase, the researchers gathered all necessary information about the current attendance system used by student organizations at NAAP.

The following were identified:

- Problems in manual attendance (errors, proxy attendance, slow reporting)
- Need for centralized event management
- Need for facial recognition verification
- Need for a QR code backup system
- Requirement for automated reports and certificate generation

Requirements Gaps Analysis

Requirements gap analysis was conducted to compare the current event attendance practices at the National Aviation Academy of the Philippines (NAAP) with the system capabilities needed for accurate, secure, and efficient event management. The analysis focused on how attendance is recorded, verified, stored, and reported during student organization events.

The analysis showed that existing procedures rely heavily on manual sign-in sheets and separate digital files, which limits automation and real-time monitoring. Attendance is based on handwritten entries or visual confirmation, making it difficult to ensure that the recorded attendee is the actual participant. This creates opportunities for proxy attendance and inaccurate records.

Another gap identified is the lack of centralized data management. Attendance records and event details are stored in different locations, resulting in delayed report generation and additional administrative work. Security is also limited, as current methods do not use standardized authentication or role-based access, increasing the risk of unauthorized access to student data.

To address these gaps, the proposed system introduces facial recognition-based attendance, centralized event and attendance records, secure user authentication, and automated reporting and certificate generation. These features directly respond to the limitations of the existing system and support more reliable, efficient, and transparent event attendance management at NAAP.

Alternative Solutions and Feasibility Study

This section presents the different solution approaches considered for the system. It compares each option and explains why the selected solution is the most suitable based on system requirements, flexibility, and overall effectiveness.

Proposed Technology Alternatives

Several technology alternatives were examined to identify the most suitable solution for improving event attendance management for student organizations at the National Aviation Academy of the Philippines (NAAP). The alternatives considered included enhancing existing survey or attendance tools, adopting commercial event management platforms, and developing a custom web-based system. Each option was evaluated based on system flexibility, integration capability, security, scalability, cost, and suitability for institutional requirements.

Enhancing existing survey and attendance tools was initially considered due to their low cost and ease of use. However, these tools are limited to basic data collection and lack real-time identity verification, centralized data management, and automated reporting features. Attendance validation still requires manual checking, making the process vulnerable to errors, proxy attendance, and delayed report generation.

Commercial event management platforms were also assessed as potential solutions. While these platforms offer pre-built features such as event registration and attendance tracking, they often function as standalone systems with limited integration with institutional databases. Customization options are also restricted, particularly for implementing facial recognition, QR-code fallback mechanisms, and institution-specific reporting requirements. In addition, subscription and licensing fees increase long-term operational costs.

The development of a custom web-based event management system was selected as the most appropriate solution. A custom system allows full integration with institutional student records, supports machine learning-based facial recognition for accurate attendance verification, and provides centralized data

storage. It also enables automated report generation, certificate issuance, and role-based access control, ensuring data security and reliability.

Overall, the custom web-based system best addresses the functional, operational, and security needs identified in Chapter 1. Its flexibility and scalability support long-term institutional use while aligning with NAAP's goal of modernizing student organization event management and attendance monitoring.

Feasibility Study

This section presents the different solution approaches considered for the system and explains why the selected solution is the most appropriate. It also evaluates the feasibility of the proposed system in terms of technical, operational, economic, and schedule aspects to determine if it can be successfully developed and implemented.

Technical Feasibility

The technical feasibility of the proposed Web-Based Event Management System with Machine learning based Facial Attendance for the Student Organization of the National Aviation Academy of the Philippines was assessed based on the availability of technological resources, system compatibility, and the capability of existing infrastructure at NAAP to support a web-based event management platform with facial recognition functionality. The assessment focused on server readiness, network connectivity, software compatibility, and the implementation requirements for machine-learning–based facial attendance.

NAAP has access to basic institutional computing resources and network infrastructure that are sufficient to support web-based applications. The proposed system is designed to operate using standard web technologies, databases, and machine learning libraries that do not require specialized or high-performance

hardware. Facial recognition processing is implemented in a controlled manner suitable for event-based attendance, reducing the need for continuous high-load computation. Attendance verification is performed only during event sessions, making the system manageable within existing server and network capacities.

Reliable internet connectivity within the institution enables real-time system access for students, organization officers, and administrators. This supports online event creation, live attendance recording, automated report generation, and certificate issuance. The inclusion of a QR code fallback mechanism further enhances technical reliability by ensuring system continuity in cases where facial recognition may be affected by lighting conditions, camera limitations, or temporary connectivity issues.

Potential technical risks include increased server usage during large-scale events and the gradual growth of stored facial and attendance data. These risks can be addressed through proper database management, data optimization, and periodic system maintenance. Based on the compatibility of required technologies with existing infrastructure and the manageable level of technical risks, the proposed system is considered technically feasible for implementation at NAAP.

Operational Feasibility

The operational feasibility of the proposed Web-Based Event Management System with Machine learning based Facial Attendance for Student Organization of the National Aviation Academy of the Philippines was evaluated by examining its alignment with existing student organization workflows, attendance procedures, and administrative practices at NAAP. The system is designed to enhance current event operations without disrupting established organizational structures or policies.

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The proposed system supports the typical processes involved in student organization events, including participant registration, attendance verification, report preparation, and certificate generation. Instead of relying on manual sign-in sheets and separate spreadsheets, organization officers can manage events through a centralized platform. Students participate in events as usual, but attendance is recorded using facial recognition, ensuring accurate identity verification while minimizing manual intervention.

Administrators and organization advisers benefit from centralized access to attendance data and event records. Automated reporting reduces the need for manual consolidation of attendance lists and verification of participant eligibility for certificates. This improves workflow efficiency, reduces processing delays, and enhances the reliability of event documentation used for evaluation and accreditation purposes.

User adoption was also considered in evaluating operational feasibility. Most students and staff are already familiar with web-based academic systems, suggesting minimal resistance to the proposed platform. Basic orientation and user guides are sufficient for system use. Although initial adjustment to facial attendance procedures may be required, the system's intuitive interface and backup QR mechanism support smooth operational transition. Overall, the system is deemed operationally feasible and capable of improving event management practices at NAAP.

Economic Feasibility

The economic feasibility of the proposed Web-Based Event Management System with Machine learning based Facial Attendance for Student Organization of the National Aviation Academy of the Philippines was analyzed by comparing the projected development and operational costs with the expected benefits to

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student organizations and the institution. The analysis focused on determining whether the system provides value through efficiency gains, cost reduction, and improved attendance accuracy.

The primary costs associated with the system involve software development, system configuration, deployment, and minimal training activities. Since the system uses web-based technologies and commonly available development tools, licensing and infrastructure expenses are kept low. The use of existing institutional hardware and internet resources further reduces implementation costs, making the project financially attainable.

Current manual attendance procedures require significant administrative effort, including manual encoding, verification, and report preparation. These processes consume time and manpower, especially during large events. By automating attendance verification, report generation, and certificate issuance, the proposed system reduces administrative workload and associated operational costs. Over time, these efficiency gains offset the initial development investment.

The system also provides several intangible benefits, including improved data accuracy, enhanced credibility of attendance records, reduced proxy attendance, and increased confidence in organizational documentation. These benefits support better decision-making, fair evaluation of student participation, and improved institutional record-keeping. Considering the relatively low development cost and the long-term operational and administrative benefits, the proposed system is considered economically feasible and sustainable for NAAP.

Schedule Feasibility

The schedule feasibility of the proposed Web-Based Event Management System with Machine Learning-Based Facial Attendance for Student Organizations of the National Aviation Academy of the Philippines (NAAP) was

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evaluated by examining the project timeline, development phases, and deployment schedule in relation to the institution's academic calendar and student organization event schedules. The purpose of this assessment was to determine whether the system could be completed and implemented within an acceptable timeframe without disrupting regular academic activities and organizational operations.

The system development follows an Agile Software Development Life Cycle, where progress is achieved through iterative sprint cycles rather than a strictly sequential process. Each sprint includes planning, development, testing, and review activities, allowing flexibility in scheduling and enabling the team to adapt to changes without disrupting the overall timeline. The process progresses through clearly defined and sequential phases, including requirements analysis, system design, development, testing, deployment, and maintenance. Each phase was assigned a specific timeframe to ensure systematic progress and proper documentation before moving to the next stage. This structured approach supports accurate scheduling and minimizes the risk of overlapping tasks and uncontrolled changes.

The implementation timeline was planned so that system development and testing are completed prior to scheduled student organization events requiring attendance monitoring. This allows sufficient time for system validation, correction of identified issues, and user orientation. Deploying the system ahead of major events ensures that students, organization officers, and administrators are familiar with system functions, reducing operational issues during actual attendance recording.

Potential schedule risks include delays in system development due to technical challenges in facial recognition integration, extended testing periods, or limited stakeholder availability for requirement validation. These risks are mitigated

through effective sprint planning, continuous backlog refinement, and regular stakeholder feedback, which are essential principles of the Agile methodology. Progress tracking through iterative sprint cycles and ongoing documentation further supports timely project completion.

User training and system orientation activities were incorporated into the deployment schedule to ensure readiness and smooth system adoption. The web-based nature of the system minimizes training complexity, enabling users to quickly understand and utilize system features. Based on the structured development phases, alignment with institutional schedules, and effective risk mitigation strategies, the proposed system is considered schedule feasible and suitable for implementation within NAAP's operational timeframe.

Cost–Benefit Analysis

Overview

This cost–benefit analysis examines the economic feasibility of developing and implementing the Web-Based Event Management System with Machine Learning–Based Facial Attendance for Student Organizations of the National Aviation Academy of the Philippines (NAAP) using the Agile Software Development Life Cycle (SDLC). The analysis compares the projected development and operational costs with the expected tangible and intangible benefits over a three-year system lifecycle, which is commonly used in academic feasibility and system evaluation studies.

The Agile model follows an iterative and incremental development process, where system features are developed, tested, and improved through repeated sprint cycles. This approach allows continuous refinement, early issue detection, and better alignment with user needs. Unlike traditional linear models, Agile

promotes flexibility and adaptability, enabling developers to respond effectively to changes in requirements throughout the development process.

By applying the Agile methodology, development costs are distributed across multiple iterations, allowing better flexibility in resource allocation and reducing risks associated with late-stage changes. This approach minimizes the likelihood of costly revisions by identifying issues early in the development cycle and ensuring that system features are continuously validated.

The expected benefits of the system, including improved attendance accuracy, reduced administrative workload, enhanced data integrity, and automated reporting and certificate generation, are anticipated to outweigh the initial development and maintenance costs over the evaluation period. Overall, the adoption of an Agile-based development approach supports both economic efficiency and system effectiveness, making the proposed system a viable and sustainable solution for NAAP.

Cost–Benefit Analysis

Phase 1: Requirements Analysis and Planning (Agile Model)

Assumptions (Labor Cost Estimates)

Role	Hourly Rate (₱)
Project Manager	700
System Analyst	600
Developer	800
UI/UX Designer	600

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QA Tester	450
Stakeholder Representative	700

Number of stakeholders involved: **5**

1. Stakeholder Identification and Engagement Cost

Activity	Role	No. of Persons	Hourly Rate (₱)	Hours	Computation	Cost (₱)
Identify Stakeholders	Project Manager	1	700	2	$1 \times 600 \times 2$	1,200
Identify Stakeholders	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Stakeholder Meeting	Project Manager	1	700	3	$1 \times 700 \times 3$	2,100
Stakeholder Meeting	Stakeholders	5	700	2	$1 \times 1,300 \times 2$	2,600

Total Cost: ₱7,700

2. Requirement Elicitation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)

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Requirement Interview	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Requirement Workshop	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱5,600

3. User Story Development Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Story Writing	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Feature Discussion	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
UI Concept Planning	UI/UX Designer	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱5,600

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4. Product Backlog Creation and Prioritization Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Backlog Organization	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Feature Prioritization	Project Manager	1	700	2	$1 \times 700 \times 3$	2,100
Technical Review	Developer	1	800	3	$1 \times 800 \times 3$	2,400

Total Cost: ₱6,300

5. Sprint Planning Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Sprint Planning Meeting	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Sprint Planning Meeting	Developer	2	1,600	3	$1 \times 1,600 \times 3$	4,800
Sprint Planning Meeting	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱8,000

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B. Total Cost of Phase 1

Activity	Cost(₱)
Stakeholder Identification and Engagement Cost	7,700
Requirement Elicitation Cost	5,600
User Story Development Cost	5,600
Product Backlog Creation and Prioritization Cost	6,300
Sprint Planning Cost	8,000

Total Cost: ₱33,200

Phase 2: System Design & Prototype Development (Agile Model)

Key activities include:

- 1.System Architecture Design
- 2.Database Design and Data Modeling
- 3.UI/UX Wireframing and Interface Design
- 4.Prototype Development
- 5.Design Review and Validation

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A. Cost Analysis

Assumptions (Labor Cost Estimates)

Role	Hourly Rate (₱)
Project Manager	700
System Analyst	600
Developer	800
UI/UX Designer	600
Stakeholder Representative	700

1. System Architecture Design Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Architecture Planning	Project Manager	1	700	3	$1 \times 700 \times 3$	2,100
Technical Architecture Design	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
System Component	Developer	1	800	3	$1 \times 800 \times 3$	2,400

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Identification							
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Total Cost: ₱6,900

2. Database Design and Data Modeling Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Data Requirement Analysis	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Database Schema Design	Developer	1	800	4	$1 \times 800 \times 4$	3,200
Entity Relationship Diagram Creation	System Analyst	1	600	3	$1 \times 600 \times 4$	2,400

Total Cost: ₱7,400

3. UI/UX Wireframing and Interface Design Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Interface Planning	UI/UX Designer	1	600	3	$1 \times 600 \times 3$	1,800
Wireframe Design	UI/UX Designer	1	600	4	$1 \times 600 \times 4$	2,400

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User Flow Mapping	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
UI Prototype Mockups	UI/UX Designer	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱7,800

4. Prototype Development Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Prototype Coding	Developer	2	800	6	$2 \times 800 \times 6$	9,600
Front-End Interface Integration	Developer	1	800	4	$1 \times 800 \times 4$	3,200
System Integration	Developer	1	800	3	$1 \times 800 \times 3$	2,400
Technical Supervision	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400

Total Cost: ₱16,600

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5. Design Review and Validation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Prototype Review Meeting	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Design Evaluation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Usability Testing	QA Specialist	1	450	3	$1 \times 450 \times 3$	1,350
Stakeholder Validation Meeting	Stakeholders	5	700	2	$5 \times 700 \times 2$	7,000

Total Cost: ₱11,550

B. Total Cost of Phase 2

Activity	Cost (₱)
System Architecture Design	₱6,900
Database Design and Data Modeling	₱7,400
UI/UX Wireframing and Interface Design	₱7,800
Prototype Development	₱16,600
Design Review and Validation	₱11,550

Total Cost : ₱50,250

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Phase 3: Development & Iterative Implementation (Agile Model)

Key activities include:

1. Sprint Development (Feature Coding)
2. Front-End and Back-End Integration
3. Continuous Testing and Debugging
4. Sprint Review and Demonstration
5. Sprint Retrospective and Improvement Planning

A. Cost Analysis

Labor Cost

Role	Hourly Rate (₱)
Project Manager	700
System Analyst	600
Developer	800
UI/UX Designer	600
QA/Test Specialist	450

1. Sprint Development (Feature Coding) Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Feature Coding	Developer	2	800	14	$2 \times 800 \times 14$	22,400

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Business Logic Implementation	Developer	1	800	10	$1 \times 800 \times 10$	8,000
Database Integration	Developer	1	800	6	$1 \times 800 \times 6$	4,800
Code Review	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Sprint Coordination	Project Manager	1	700	4	$1 \times 700 \times 4$	2,800

Total Cost: ₱40,400

2. Front-End and Back-End Integration Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
API Integration	Developer	1	800	8	$1 \times 800 \times 8$	6,400
Front-End Integration	Developer	1	800	6	$1 \times 800 \times 6$	4,800
Interface Adjustment	UI/UX Designer	1	600	5	$1 \times 600 \times 5$	3,000
System Logic Verification	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400

Total Cost: ₱16,600

3. Continuous Testing and Debugging Cost

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Unit Testing	Developer	1	800	6	$1 \times 800 \times 6$	4,800
System Testing	QA Specialist	1	450	8	$1 \times 450 \times 8$	3,600
Bug Fixing	Developer	1	800	6	$1 \times 800 \times 6$	4,800

Total Cost: P13,200

4. Sprint Review and Demonstration Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Sprint Review Meeting	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
System Demonstration	Developer	1	800	3	$1 \times 800 \times 3$	2,400
Stakeholder Review	Stakeholders	5	700	2	$5 \times 700 \times 2$	7,000

Total Cost: P10,800

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5. Sprint Retrospective and Improvement Planning Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Retrospective Meeting	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Process Evaluation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
QA Feedback Analysis	QA Specialist	1	450	3	$1 \times 450 \times 3$	1,350

Total Cost: ₱4,550

B. Total Cost of Phase 3

Activity	Cost (₱)
Sprint Development (Feature Coding)	40,400
Front-End and Back-End Integration	16,600
Continuous Testing and Debugging	13,200
Sprint Review and Demonstration	10,800
Sprint Retrospective and Improvement Planning	4,550

Total Cost : ₱85,550

Phase 4: System Testing, Deployment, and Maintenance (Agile Model)

Key activities include:

1. System Testing and Quality Assurance
2. User Acceptance Testing (UAT)
3. System Deployment and Configuration
4. User Training and Documentation
5. Initial System Maintenance and Support

A. Cost Analysis

Assumptions (Labor Cost Estimates)

Role	Hourly Rate (₱)
Project Manager	700
System Analyst	600
Developer	800
UI/UX Designer	600
QA/Test Specialist	450
Stakeholder Representative	700

Number of stakeholders involved: 5

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1. System Testing and Quality Assurance Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Test Plan Preparation	QA Specialist	1	450	4	$1 \times 450 \times 4$	1,800
Functional Testing	QA Specialist	1	450	8	$1 \times 450 \times 8$	3,600
Integration Testing	Developer	1	800	5	$1 \times 800 \times 5$	4,000
Performance Testing	Developer	1	800	4	$1 \times 800 \times 4$	3,200
Bug Fix Implementation	Developer	1	600	5	$1 \times 600 \times 5$	3,200

Total Cost: ₱ 15,600

2. User Acceptance Testing (UAT) Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
UAT Planning	Project Manager	1	700	3	$1 \times 700 \times 3$	2,100
UAT Coordination	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400

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User Acceptance Testing Session	Stakeholders	5	700	4	$5 \times 700 \times 4$	14,000
Issue Identification	QA Specialist	1	450	3	$1 \times 450 \times 3$	1,350
UAT Report Preparation	System Analyst	1	650	2	$1 \times 650 \times 2$	1,300

Total Cost: ₱21,150

3. System Deployment and Configuration Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Deployment Planning	Project Manager	1	700	3	$1 \times 700 \times 3$	2,100
Server Configuration	Developer	1	800	5	$1 \times 800 \times 5$	4,000
Database Deployment	Developer	1	800	4	$1 \times 800 \times 4$	3,200
System Configuration	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Deployment	QA	1	450	3	$1 \times 450 \times 3$	1,350

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Verification	Specialist					
Hosting / Cloud Setup	Cloud Services	—	—	—	Estimated	1,000

Total Cost: ₱14,050

4. User Training and Documentation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Training Preparation	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
UI/UX Guide Creation	UI/UX Designer	1	600	4	$1 \times 600 \times 4$	2,400
User Training Session	Project Manager	1	700	3	$1 \times 700 \times 3$	2,100
User Training Session	Stakeholders	5	700	3	$5 \times 700 \times 3$	10,500
User Manual Documentation	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400

Total Cost: ₱19,800

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5. Initial System Maintenance and Support Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Monitoring	Developer	1	800	4	$1 \times 800 \times 4$	3,200
Bug Fix and Patch Updates	Developer	1	800	5	$1 \times 800 \times 5$	4,000
User Support Assistance	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Performance Monitoring	QA Specialist	1	450	3	$1 \times 450 \times 3$	1,350
Maintenance Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱ 12,750

B. Total Cost of Phase 4

Activity	Cost (₱)
System Testing and Quality Assurance	15,600
User Acceptance Testing	21,150
System Deployment and Configuration	14,050
User Training and Documentation	19,800

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Initial System Maintenance and Support	12,750
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Total Cost : ₱82,350

Phase 5: System Evaluation and Continuous Improvement (Agile Model)

Key activities include:

1. System Performance Evaluation
2. User Feedback Collection and Analysis
3. System Enhancement Planning
4. Security and Performance Review
5. Continuous Improvement Implementation

A. Cost Analysis

Assumptions (Labor Cost Estimates)

Role	Hourly Rate (₱)
Project Manager	700
System Analyst	600
Developer	800
UI/UX Designer	600
QA/Test Specialist	450

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Stakeholder Representative	700
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Number of stakeholders involved: 5

1. System Performance Evaluation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Performance Monitoring	Developer	1	800	5	$1 \times 800 \times 5$	4,000
System Usage Analysis	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Performance Testing	QA Specialist	1	450	4	$1 \times 450 \times 4$	1,800
Performance Report Preparation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Evaluation Meeting	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400

Total Cost: ₱11,400

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2. User Feedback Collection and Analysis Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Survey Preparation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Feedback Collection	Stakeholders	5	700	2	$5 \times 700 \times 2$	7,000
Feedback Analysis	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Result Presentation	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱14,400

3. System Enhancement Planning Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Requirement Identification	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400

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Improvement Planning	Project Manager	1	700	3	$1 \times 700 \times 3$	2,100
Technical Feasibility	Developer	1	800	4	$1 \times 800 \times 4$	3,200
UI/UX Enhancement Planning	UI/UX Designer	1	600	3	$1 \times 600 \times 3$	1,800
Enhancement Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱11,300

4. Security and Performance Review Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Security Assessment	Developer	1	800	4	$1 \times 800 \times 4$	3,200
Vulnerability Testing	QA Specialist	1	450	4	$1 \times 450 \times 4$	1,800
System Audit Review	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

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Security Review Meeting	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Security Documentation	System Analyst	1	600	3	$1 \times 600 \times 2$	1,200

Total Cost: ₱9,400

5. Continuous Improvement Implementation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Minor System Enhancements	Developer	1	800	5	$1 \times 800 \times 5$	4,000
UI/UX Improvements	UI/UX Designer	1	600	4	$1 \times 600 \times 4$	2,400
Regression Testing	QA Specialist	1	450	4	$1 \times 450 \times 4$	1,800
Enhancement Review	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Improvement Documentation	System Analyst	1	600	2	$1 \times 600 \times 2$	1,200

Total Cost: ₱11,200

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B. Total Cost of Phase 5

Activity	Cost (₱)
System Performance Evaluation	11,400
User Feedback Collection and Analysis	14,400
System Enhancement Planning	11,300
Security and Performance Review	9,400
Continuous Improvement Implementation	11,200

Total Cost : ₱57,700

Phase 6: Long-Term Maintenance and System Sustainability (Agile Model)

Key activities include:

1. System Monitoring and Maintenance
2. Technical Support and Issue Resolution
3. System Updates and Feature Enhancements
4. Security Monitoring and Risk Management
5. System Documentation and Knowledge Transfer

A. Cost Analysis

Assumptions (Labor Cost Estimates)

Role	Hourly Rate (₱)
Project Manager	700

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System Analyst	600
Developer	800
UI/UX Designer	600
Stakeholder Representative	450

1. System Monitoring and Maintenance Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Performance	Developer	1	800	5	$1 \times 800 \times 5$	4,000
Database Maintenance	Developer	1	800	4	$1 \times 800 \times 4$	3,200
Log Monitoring and Analysis	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Maintenance Planning	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Maintenance Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱12,800

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2. Technical Support and Issue Resolution Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Support Assistance	Technical Support Staff	1	250	5	$1 \times 250 \times 5$	1,250
Issue Diagnosis	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Bug Fix Implementation	Developer	1	800	4	$1 \times 800 \times 4$	3,200
System Verification	QA Specialist	1	450	3	$1 \times 450 \times 4$	1,800
Resolution Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1800

Total Cost: ₱10,650

3. System Updates and Feature Enhancements Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Enhancement Review	System	1	600	4	$1 \times 600 \times 4$	2,400

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	Analyst					
Feature Development	Developer	1	800	6	$1 \times 800 \times 6$	4,800
Interface Improvements	UI/UX Designer	1	600	4	$1 \times 600 \times 4$	2,400
Enhancement Testing	QA Specialist	1	450	4	$1 \times 450 \times 4$	1,800
Update Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱ 13,200

4. Security Monitoring and Risk Management Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Security Monitoring	Developer	1	800	4	$1 \times 800 \times 4$	3,200
Vulnerability Assessment	QA Specialist	1	450	4	$1 \times 450 \times 4$	1,800
Risk Assessment	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

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Security Review Meeting	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Security Report Documentation	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800

Total Cost: ₱10,000

5. System Documentation and Knowledge Transfer Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Technical Documentation Update	System Analyst	1	600	4	$1 \times 600 \times 4$	2,400
Maintenance Guide	System Analyst	1	600	3	$1 \times 600 \times 3$	1,800
Knowledge Transfer	Project Manager	1	700	2	$1 \times 700 \times 2$	1,400
Staff Orientation	Technical Support Staff	2	250	3	$1 \times 250 \times 3$	750
Documentation Archiving	System Analyst	1	600	2	$1 \times 600 \times 2$	1,200

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Total Cost: ₱7,550

B. Total Cost of Phase 6

Activity	Cost (₱)
System Monitoring and Maintenance	12,800
Technical Support and Issue Resolution	10,650
System Updates and Feature Enhancements	13,200
Security Monitoring and Risk Management	10,000
System Documentation and Knowledge Transfer	7,550

Total Cost : ₱54,200

Agile Development Cost Summary

Phase	Description	Total Cost (₱)
Phase 1	Requirements Analysis and Planning	33,200
Phase 2	System Design and Prototype Development	50,250
Phase 3	Development and Iterative Implementation	85,550
Phase 4	System Testing, Deployment, and Maintenance	82,350
Phase 5	System Evaluation and Continuous Improvement	57,700

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Phase 6	Long-Term Maintenance and System Sustainability	54,200
Total Project Cost	Overall Agile System Development Cost	₱363,250

Development Cost

Cost Component	Description	Estimated Cost
Requirement Analysis & Planning	Stakeholder consultation, requirements gathering, and project planning activities	35,000
System Design & Architecture	UI/UX design, database design, system architecture planning, and technical specifications	50,000
Development (Core Modules)	Coding of core system features including facial recognition integration, QR attendance module, database integration, and backend/frontend development	160,000
Testing & Evaluation	System testing, debugging, quality assurance, user acceptance testing (UAT), and performance evaluation	45,000

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Total Development Cost		290,000
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Operational and Maintenance Costs (Annual)

Cost Component	Estimated Annual Cost
Cloud Hosting / Server	₱12,000
System Maintenance	₱20,000
Technical Support	₱15,000
System Updates and Security Monitoring	₱10,000
Total Annual Operational Cost	57,000 PHP

Total Cost Over 3 Years

Total Cost = Development Cost+(3×Annual Operational Cost)

Total Cost = 290,000+(3x57,000) = 461,000 PHP

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Benefit Analysis

1. Tangible Benefits (Annual)

Benefit	Description	Estimated Savings Per Year
Reduced Paper and Printing Costs	Minimizes the use of paper for attendance sheets, reports, and documentation through digital record-keeping	₱10,000
Reduced Manual Attendance Processing	Eliminates time-consuming manual encoding and checking of attendance through automated facial recognition and QR scanning	₱25,000
Reduced Event Documentation Processing Time	Speeds up report generation and attendance tracking, reducing time spent on organizing and verifying event records	₱20,000
Reduced Administrative Workload	Decreases staff effort in managing attendance, monitoring participation, and preparing reports	₱15,000

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Total Tangible Benefits (Annual)		70,000 PHP
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Intangible Benefits

- Enhanced Attendance Integrity and Credibility
- Strengthening Data Privacy and Legal Compliance
- Improved Stakeholder Satisfaction
- High Interaction Capability and Accessibility
- Reliability: Fault Tolerance via QR Fallback

Total Benefits Over 3 Years

Total Benefits = 3×70,000 = **210,000 PHP**

Net Benefit Computation

Net Benefit=Total Benefits–Total Costs

Net Benefit = 210,000 - 461,000 = **251,000 PHP.**

Return on Investment (ROI)

$$ROI = \left(\frac{\text{Net Benefit}}{\text{Total Cost}} \right) \times 100$$

$$ROI = \left(\frac{251,000}{461,000} \right) \times 100 = 54.45\%$$

Agile-Specific Cost Advantage Justification

Using Agile SDLC provides the following economic advantages:

- Early feature validation reduces development waste through continuous stakeholder feedback and iterative refinement
- Continuous testing lowers post-deployment maintenance cost by identifying and fixing defects early in each iteration
- Incremental delivery allows partial system functionality to be usable before full project completion
- Reduced risk of system rejection by stakeholders due to continuous involvement and iterative approval of outputs
- Faster realization of operational benefits through early delivery of functional system components

Cost Benefit Summary

Metric	Value
Total Development Cost	290,000
Total 3-Year Operational Cost	171,000
Total System Cost (3 Years)	461,000

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Total Benefits (3Years)	210,000
Net Benefit	251,000
ROI	54.45%

The Agile Software Development Life Cycle was selected for this project because it offers significant cost advantages throughout the system development and implementation process.

Agile focuses on iterative and incremental development, allowing the project team to build the system in smaller phases or sprints rather than completing the entire system at once. This approach helps reduce development risks and prevents large financial losses if changes or adjustments are needed during the project.

Additionally, Agile encourages continuous testing, stakeholder feedback, and regular reviews, which allow issues and system defects to be identified early in the development process. Detecting problems early helps minimize the cost of fixing errors compared to traditional development models where issues are often discovered later.

Agile also improves resource utilization by promoting collaboration among developers, analysts, designers, and stakeholders, ensuring that development efforts are aligned with the actual needs of the users.

Furthermore, Agile allows the system to deliver functional components earlier, enabling the organization to start benefiting from the system while further improvements are still being developed. Because of these advantages, Agile helps

optimize development costs while ensuring flexibility, efficiency, and continuous improvement throughout the system lifecycle.

System Requirements Specification

The System Requirements Specification (SRS) defines the overall requirements of the Web-Based Event Management and Facial Recognition Attendance System and serves as a guide for its design, development, testing, and implementation. Its purpose is to clearly describe the system's expected functions, performance standards, and operating conditions to ensure that it meets the needs of students, organization officers, and administrators.

The SRS acts as a technical reference that outlines how the system should function, the features it must provide, and the conditions under which it should operate reliably. By establishing clear and structured requirements, it supports consistent system development, strengthens security and efficiency, and ensures accurate attendance monitoring, centralized event management, and dependable data processing within the institutional environment.

Stakeholder Identification & User Roles

The proposed Web-Based Event Management System with Machine Learning-Based Facial Recognition Attendance supports multiple stakeholder groups whose responsibilities are essential to the effective implementation and operation of student organization events at the National Aviation Academy of the Philippines (NAAP). The identification of stakeholder roles ensures that system functionalities align with organizational workflows and support role-based access control to maintain attendance accuracy, data security, accountability, and system reliability.

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Students serve as primary participants responsible for attending student organization events and recording their attendance using the system's facial recognition technology. Through the system, students log in using authorized credentials, participate in registered events, and verify their attendance through real-time facial recognition, with a QR-code fallback mechanism available when necessary. Integration of student access with official student records ensures that only authorized and enrolled students can participate in events, thereby improving attendance validity and preventing proxy or duplicate submissions.

Student organization officers participate in the event management process by creating, organizing, and monitoring events within the system. Officers are responsible for encoding event details, managing participant lists, monitoring real-time attendance records, and generating attendance reports. The centralized platform enables officers to efficiently manage event data, replacing manual sign-in sheets and fragmented spreadsheets while improving operational efficiency and documentation accuracy.

Institutional administrators oversee broader event monitoring and system usage within the academy. They utilize system-generated dashboards and consolidated attendance reports to track participation trends, verify event compliance, and support documentation requirements for organizational accreditation and reporting purposes. Administrative oversight strengthens accountability and ensures that attendance records are reliable and properly maintained.

System administrators manage overall system operations, including user account management, event configuration settings, facial recognition database maintenance, QR-code generation, system security monitoring, and data backup. Administrators ensure that access permissions are properly implemented and that

attendance data are securely stored and processed in compliance with institutional data privacy policies and national data protection regulations.

The integration of clearly defined stakeholder roles enhances system usability, supports institutional accountability, and strengthens the reliability of attendance records. Role-based system access ensures that users interact only with functions relevant to their responsibilities, thereby improving operational efficiency, preventing unauthorized access, reducing attendance errors, and reinforcing institutional governance in student organization event management.

System Administrators

The Web-Based Event Management System with Machine Learning–Based Facial Recognition Attendance is designed to provide important system functions that support automated, secure, and centralized event management at the National Aviation Academy of the Philippines (NAAP). The functional requirements explain how the system works, how users interact with it, and how it helps improve attendance accuracy, event documentation, and overall efficiency.

The system shall provide a secure event registration and attendance module that allows students to join events and record their attendance using a web-based platform. The system shall check student eligibility by connecting user accounts to official student records. This ensures that only authorized and enrolled students can participate. The system shall also prevent duplicate attendance and unauthorized entries through automatic validation.

The system shall require users to log in using their official school credentials. It shall use role-based access control, meaning users can only access features based on their assigned role, such as student, organization officer, or administrator. This helps protect attendance records and prevent unauthorized access or changes.

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The system shall include a facial recognition feature powered by machine learning. This feature captures a student's face in real time and matches it with stored images in the database to confirm identity before recording attendance. This helps reduce proxy attendance and improves accuracy. If facial recognition cannot be used due to technical or environmental issues, the system shall provide a QR-code fallback option to ensure attendance can still be recorded properly.

The system shall automatically generate attendance reports and participation summaries for each event. These reports shall be available to organization officers and authorized administrators to help them monitor attendance and prepare documentation when needed.

The system shall also generate certificates automatically for students who meet the required attendance or participation criteria. Students can download their certificates directly from the system once eligibility is verified.

The system shall provide administrative functions that allow system administrators to manage user accounts, create and configure events, maintain facial recognition data, generate QR codes, monitor system activity, and perform data backup. These controls help ensure system reliability, data security, and proper system maintenance.

By implementing these functional requirements, the system improves attendance accuracy, reduces manual work, and ensures secure and organized event documentation. Through the integration of facial recognition, QR-code backup, automated reporting, and certificate generation in one centralized platform, the system enhances efficiency and strengthens event management processes at NAAP.

Functional Requirements

The Web-Based Event Management System with Machine Learning–Based Facial Attendance is designed to provide a centralized, secure, and automated platform for managing student organization events at the National Aviation Academy of the Philippines (NAAP). The functional requirements define the system’s behaviors, user interactions, and operational processes to ensure accurate attendance monitoring, efficient event management, and reliable documentation.

The system shall provide secure event management modules that allow organization officers and administrators to create, edit, and manage events through a web-based dashboard. The system shall enable the input of event details such as event title, date, time, venue, participant list, and attendance requirements. The system shall store all event records in a centralized database to prevent data duplication and loss.

The system shall implement authentication and authorization mechanisms to control user access based on assigned roles (Student, Organization Officer, Administrator). Users shall log in using institution-d credentials. Role-based access control shall restrict system functionality according to user responsibilities, ensuring data confidentiality and preventing unauthorized access to sensitive records.

The system shall provide a machine learning–based facial recognition attendance module that captures real-time facial images and matches them against registered student records. The system shall validate student identity before marking attendance to prevent proxy attendance. Attendance marking shall occur only after successful identity verification.

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The system shall include a QR code fallback mechanism in cases where facial recognition fails due to environmental factors such as poor lighting or unstable internet connection. The QR code system shall verify the student's identity using a unique generated code linked to their registered account. All fallback attendance logs shall be recorded and flagged accordingly for monitoring purposes.

The system shall prevent duplicate attendance entries through automated validation and submission tracking. Once attendance is successfully recorded for a specific event session, the system shall block repeated submissions from the same user.

The system shall generate automated attendance reports and event summaries that include total participants, verified attendance records, attendance percentages, and flagged discrepancies. These reports shall be accessible to authorized stakeholders such as organization officers and administrators to support monitoring and decision-making.

The system shall provide automated certificate generation for qualified participants based on predefined attendance criteria. Certificates shall be downloadable in digital format, and the system shall maintain issuance records for documentation and verification purposes.

The system shall include performance monitoring features that track facial recognition accuracy, response time, and fallback usage frequency. These logs shall support system evaluation and technical reliability assessment.

The system shall include administrative management functions that allow system administrators to manage user accounts, update student records, configure system settings, monitor activity logs, and maintain the facial recognition

database. These controls shall ensure data integrity, system reliability, and compliance with institutional policies.

The implementation of these functional requirements supports accurate attendance verification, efficient event documentation, secure data handling, and automated certificate processing. By integrating event management and biometric authentication into a centralized web-based platform, the system improves operational efficiency and strengthens attendance governance for student organizations at NAAP.

Non-Functional Requirements (ISO/IEC 25010 based)

The non-functional requirements of the Web-Based Event Management and Facial Recognition Attendance System are based on the ISO/IEC 25010 software quality model to ensure that the system meets established standards for performance, usability, security, and reliability. These requirements define how well the system performs its functions and the quality conditions under which it must operate to satisfy user expectations and institutional needs.

In terms of functional suitability, the system must accurately perform attendance verification, event management, and report generation according to user inputs and system logic. For performance efficiency, the system should respond quickly during login, facial recognition processing, and data retrieval, even when multiple users access it simultaneously. Compatibility requires the system to function correctly across different devices and modern web browsers without affecting performance or usability.

The system must also demonstrate strong interaction capability by providing an intuitive interface that allows students, organization officers, and administrators to navigate features easily. For reliability, the platform should operate consistently with minimal errors or downtime and must preserve stored data even during

unexpected interruptions. Security requirements ensure that user data is protected through authentication, access control, encryption, and audit logging mechanisms.

Regarding maintainability, the system should be structured in a modular and organized manner so updates, debugging, and improvements can be performed efficiently. Flexibility requires the system to adapt to different event types, organizational needs, and future feature enhancements. Lastly, safety considerations ensure that system operations do not compromise data integrity or user privacy and that safeguards are in place to prevent accidental data loss or unauthorized modifications.

Together, these non-functional requirements ensure that the system operates not only correctly but also efficiently, securely, and reliably, providing a high-quality platform for event management and attendance monitoring within the institution.

Proposed System Architecture and Design

This section presents the overall structure and design of the proposed system. It describes how the system components are organized and how they interact with each other through diagrams such as architecture diagrams, data flow diagrams, and process flows. These illustrations provide a clear understanding of how the system operates and supports its functionalities.

System Architecture Overview

The proposed system adopts a three-tier client-server architecture that separates the system into the Presentation Layer, Application Layer, and Database Layer to ensure scalability, security, and efficient data management.

Process Modeling and System Workflow

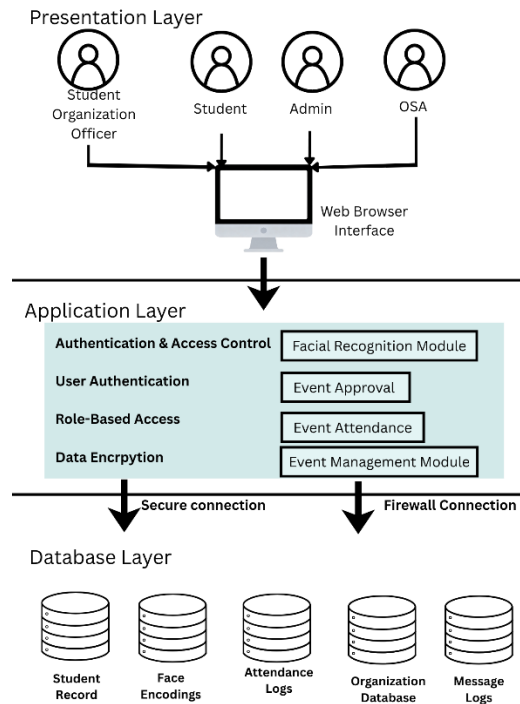


Figure 3: System Architecture of the Web-Based Event Management System with Machine Learning-Based Facial Attendance and QR Fallback for the National Aviation Academy of the Philippines.

At the Presentation Layer, the system is accessed by three primary users: students, student organization officers, and the Office of Student Affairs (OSA). These users interact with the system through a web browser interface, which provides a user-friendly platform for managing events, attendance tracking, and communication. This layer ensures accessibility across multiple devices while maintaining ease of use.

The Application Layer serves as the core of the system, handling all business logic and processing. It includes key components such as authentication

and access control, which ensures secure login and user verification. The system implements role-based access control (RBAC) to restrict access based on user roles, and data encryption to protect sensitive information. Additionally, this layer contains functional modules such as the Facial Recognition Module for automated attendance tracking, the Event Approval module for managing event requests, the Event Attendance module for recording participant presence, and the Event Management module for organizing and overseeing events. These modules work together to streamline operations and improve efficiency in managing student activities.

The Database Layer is responsible for storing and managing system data using a structured database. It consists of multiple data tables, including student records, face encodings, attendance logs, organization data, and message logs. This layer ensures efficient data retrieval, consistency, and long-term storage of records. Communication between the application and database is secured through secure connections and firewall protection, ensuring data integrity and compliance with security standards.

System Components and Modules

The proposed Web-Based Event Management System with Machine Learning–Based Facial Attendance consists of interconnected functional modules designed to support automated event management, secure attendance verification, and centralized data handling at the National Aviation Academy of the Philippines (NAAP). The modular design ensures system scalability, maintainability, and proper enforcement of role-based access among Students, Student Organization Officers, OSA, and System Administrators.

The Access Control and Authentication Module serves as the primary security component of the system. It manages user login credentials and enforces

Role-Based Access Control (RBAC) to differentiate access levels among users. This module ensures that only authorized users can perform specific actions such as event creation, attendance monitoring, and report generation, thereby maintaining system security and data integrity.

The User Profile and Records Management Module is responsible for storing and maintaining user information, including student details, organization membership, and administrative records. This module allows authorized personnel to register, update, and verify user data, ensuring that only valid users can access the system and participate in events.

The Event Management Module enables student organization officers and administrators to create, update, and manage event details. It includes features such as event scheduling, participant registration, announcements, and activity monitoring. This module ensures that all event-related information is centralized and easily accessible.

The Attendance Management Module handles the recording and verification of attendance during events. It integrates machine learning-based facial recognition using Python to automatically identify participants. The system captures facial data, performs verification, and records attendance in real time. A QR code fallback mechanism is also provided to ensure reliability in cases where facial recognition is not successful.

The Certificate Generation Module automates the creation and distribution of certificates for event participants. It verifies attendance records before generating certificates, ensuring that only eligible participants receive them. This reduces manual work and improves accuracy in certificate issuance.

The Announcement and Notification Module allows organizations and administrators to send event updates, reminders, and important notices to users.

This ensures effective communication and keeps participants informed about upcoming events and activities.

The Reporting and Analytics Module processes attendance and event data to generate reports and summaries. It provides insights such as participation rates, attendance trends, and event performance, supporting decision-making and improving organizational efficiency.

The integration of these modules creates a unified and automated system that enhances event management and attendance monitoring. The modular architecture allows easy maintenance and future improvements, such as upgrading the facial recognition component or expanding reporting features, without affecting the overall system structure.

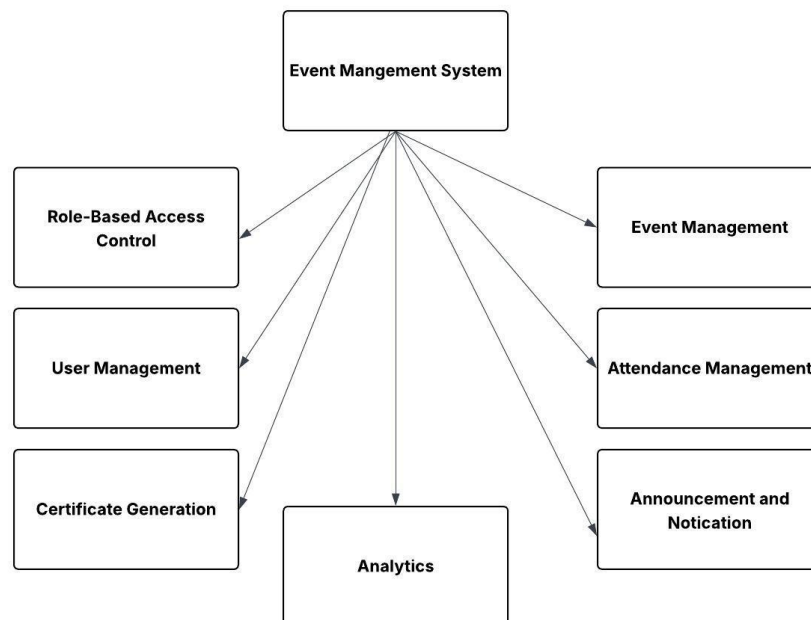


Figure 4: Improved Modular Architecture of the Web-Based Event Management System with Machine learning based Facial Attendance for Student Organization of the National Aviation Academy of the Philippines

Figure 4 presents the modular architecture of the proposed Web-Based Event Management System with Machine Learning–Based Facial Attendance. The Event Management System serves as the central processing component responsible for integrating and coordinating all system operations, including event handling, attendance tracking, and data processing. Supporting modules include the Role-Based Access Control Module, which ensures secure authentication and role-based authorization for different users such as students, student organization officers, OSA personnel, and administrators; the User Management Module, which handles user registration, profile management, and account maintenance; the Event Management Module, which manages the creation, updating, and monitoring of events; the Attendance Management Module, which records and verifies attendance using facial recognition with QR code fallback; the Certificate Generation Module, which automates the issuance of certificates for qualified participants; the Announcement and Notification Module, which facilitates communication and dissemination of information; and the Analytics Module, which processes system data to generate reports and insights for decision-making. The modular architecture promotes scalability, maintainability, and efficient integration of system components, ensuring reliable and organized event management operations.

Process Modeling and System Workflow

This section explains the structural and operational design of the proposed Web-Based Event Management System with Machine Learning–Based Facial Attendance. The system architecture is presented through analytical models such as the Context Diagram, Data Flow Diagrams (DFD Level 0 and Level 1), Entity Relationship Diagram, and procedural workflows to show how information moves across different system components.

These models illustrate how user data, event details, attendance records, certificates, announcements, and reports are processed through connected system modules. The workflow begins with user authentication and role-based access, then continues through event creation, event registration, attendance verification using facial recognition with QR code fallback, report generation, and certificate issuance.

The interaction among Students, Student Organization Officers, OSA, and Administrators is clearly represented in these models, highlighting their roles in event participation, event management, monitoring, and documentation. By mapping each stage of data processing, from initial user access to final reporting and certificate generation, this section demonstrates how the system improves efficiency, ensures accurate attendance verification, and supports organized event management within NAAP.

Context Diagram

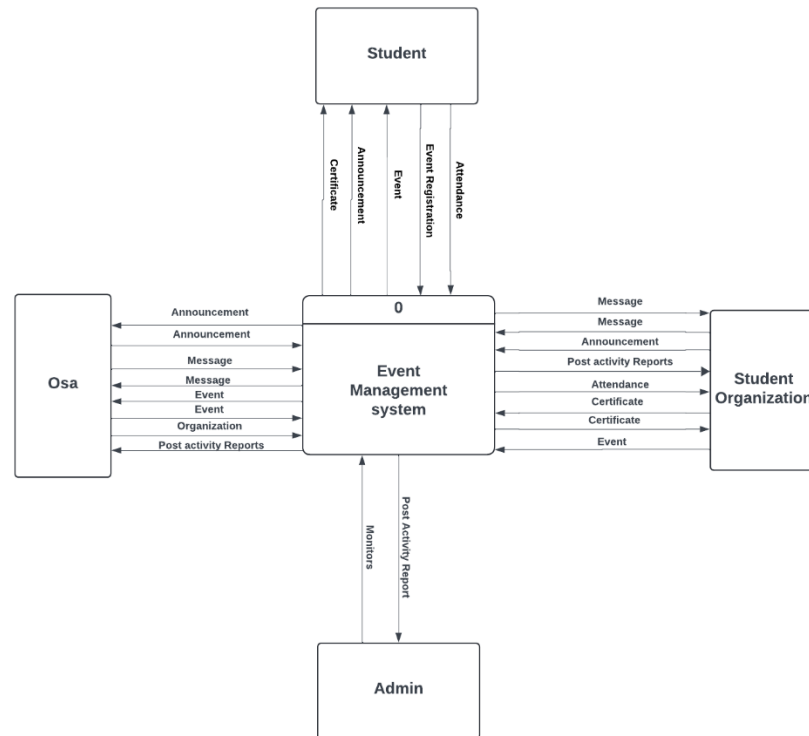


Figure 5: Context Diagram of Web-Based Event Management System with Machine learning based Facial Attendance for Student Organization of the National Aviation Academy of the Philippines

Figure 5 presents the Context Diagram of the proposed Event Management System. The diagram illustrates the interaction between the system and its external entities, namely the Student, Student Organization, Office of Student Affairs (OSA), and Administrator. It shows the flow of data such as event details, announcements, attendance records, certificates, and reports exchanged between users and the system. The diagram highlights the system as a centralized platform that processes and manages all event-related activities while supporting communication and coordination among stakeholders.

Data Flow Diagram (Level 0)

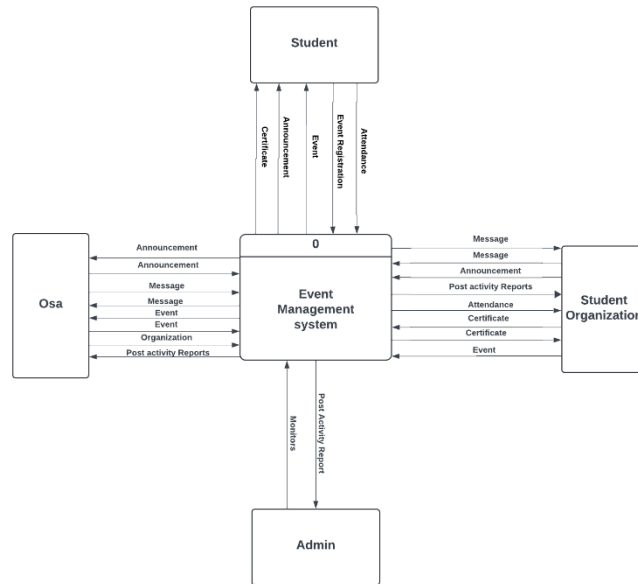


Figure 6: Data Flow Diagram Level 0 of the Web-Based Event Management System

Figure 6 presents the Level 0 Data Flow Diagram of the proposed Event Management System. The diagram illustrates the major processes of the system, including event management, attendance tracking, certificate generation, reporting, and communication. It shows how data flows between external entities such as Students, Student Organizations, OSA, and Administrator, and the system's central processes. The diagram also highlights the interaction with key data stores such as user records, event data, attendance data, and reports. Overall, the Level 0 DFD provides a high-level overview of how information is processed within the system.

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Data Flow Diagram (Level 1)

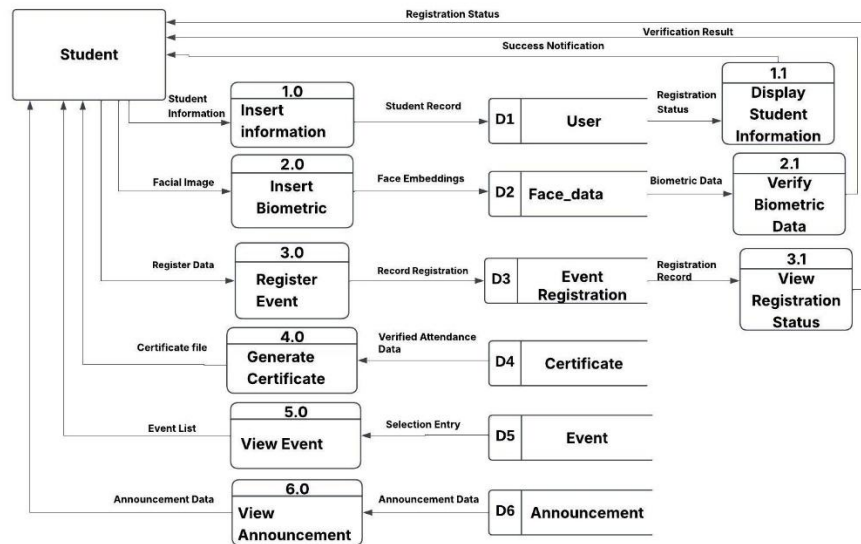


Figure 7: Dat Flow Level 1 of Students

Figure 7 presents the Level 1 Data Flow Diagram for the Student module of the proposed system. The diagram illustrates the processes involved in student interaction, including viewing events, registering for events, checking attendance records, receiving announcements, and accessing certificates. It also shows how student data is exchanged with system data stores such as Event, Attendance, Certificate, and Announcement. The diagram highlights the role of students as primary users who participate in events and access system-generated information.

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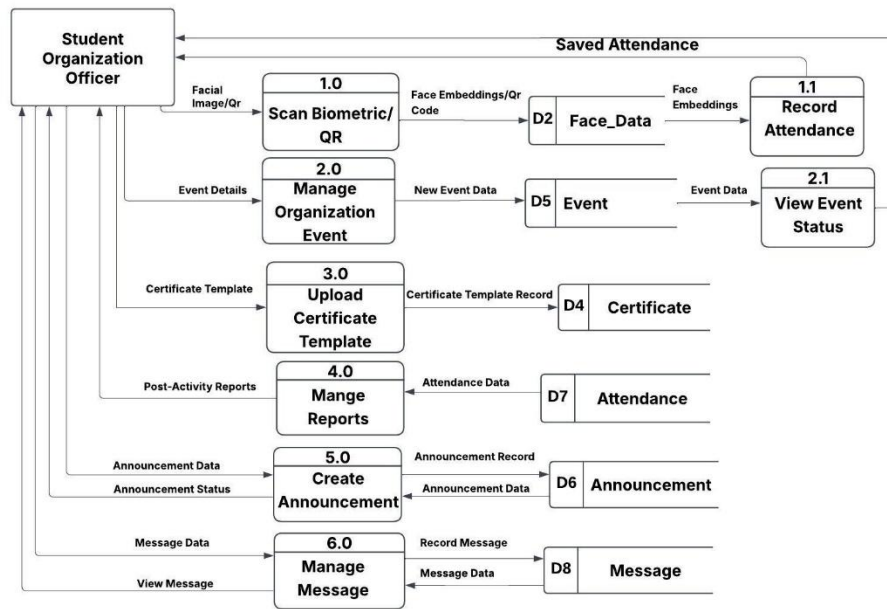


Figure 8: Data Flow Level 1 of Student Organization Officer

Figure 8 describes how Student Organization Officers interact with the system to manage events and related activities. The Scan Biometric/QR process (1.0) captures attendance using facial recognition or QR codes, which are stored in the Face_Data database. This ensures accurate and automated attendance tracking. The Record Attendance process (1.1) processes the captured data and saves attendance records, reducing manual errors. The Manage Organization Event process (2.0) allows organizations to create and update event details, which are stored in the event database. The View Event Status process (2.1) enables monitoring of event progress and participation. The Upload Certificate Template process (3.0) allows organizations to prepare certificates that will later be issued to participants. The Manage Reports process (4.0) uses attendance data to generate reports for evaluation. The Create Announcement process (5.0) and Manage Message process (6.0) allow communication between users. Figure 8

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shows that student organizations are responsible for event execution, attendance tracking, and communication, making them key operational users of the system.

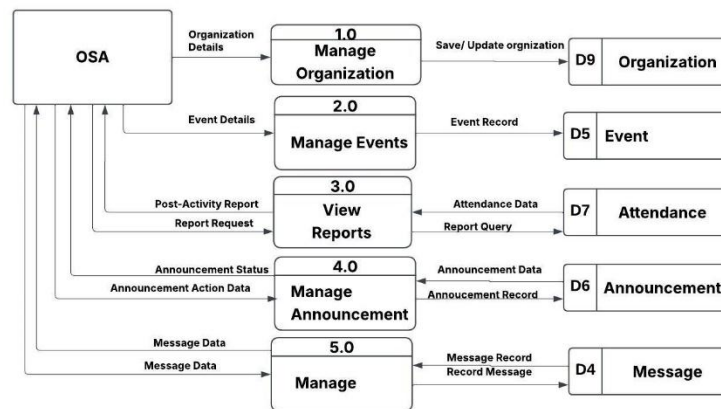


Figure 9: Data Flow Level 1 of OSA

Figure 9 shows the role of the Office of Student Affairs as the supervisory body in the system. The Manage Organization process (1.0) allows OSA to approve and update organization records, ensuring only legitimate organizations operate in the system. The Manage Events process (2.0) enables OSA to oversee all events, ensuring compliance with institutional policies. The View Reports process (3.0) retrieves attendance data to generate reports, helping in decision-making and evaluation of activities. The Manage Announcement process (4.0) ensures that announcements are appropriate and properly disseminated. The Manage Message process (5.0) supports communication between OSA and other users. Figure 9 highlights that OSA functions as a regulatory and monitoring authority, ensuring proper system usage and event management.

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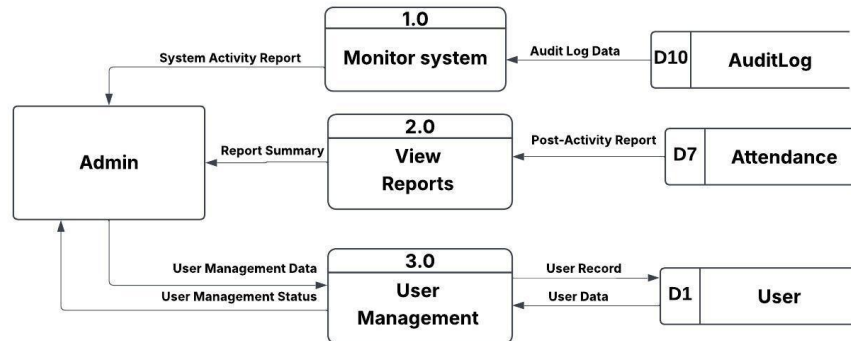


Figure 10: Data Flow Level 1 of Admin

The Figure 10 Admin DFD illustrates the key processes performed by the Administrator in managing and monitoring the system. The Monitor System process (1.0) allows the administrator to track system activities using audit logs. This ensures that all user actions are recorded and can be reviewed for security and accountability purposes. The View Reports process (2.0) enables the administrator to access attendance data and generate summarized reports. These reports help in evaluating event participation and system performance. The User Management process (3.0) allows the administrator to manage user records, including adding, updating, and maintaining user information stored in the database. Figure 10 highlights that the Administrator plays a critical role in ensuring system security, data accuracy, and operational efficiency.

Procedural Workflow

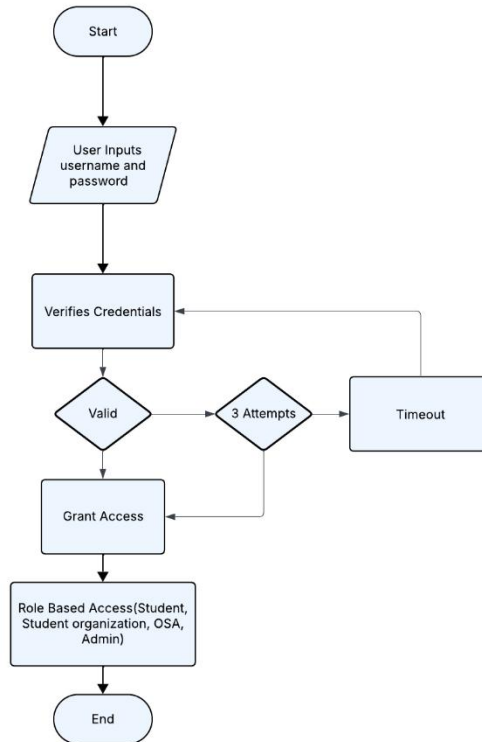


Figure 11: Procedural workflow of Login

Figure 11 presents the Procedural Workflow of the Login Module. The diagram illustrates the step-by-step process of user authentication, starting from entering login credentials to system verification. It includes decision points for validating credentials, limiting login attempts, and handling timeout conditions. Upon successful authentication, the system grants access and assigns role-based permissions. This workflow ensures secure system access and protects user data from unauthorized entry.

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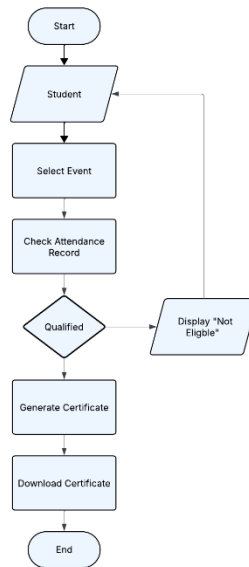


Figure 12: Procedural workflow of Certificate Generation Module

Figure 12 presents the process of generating certificates for students who participate in events. The process begins when a student selects a specific event within the system. The system then verifies the student's attendance record by checking whether the required participation criteria have been met. This includes confirming that the student has successfully attended or completed the event. If the student meets the required conditions, the system automatically generates a digital certificate and provides an option for the student to download or access it. However, if the student does not satisfy the attendance requirements, the system displays a "Not Eligible" message, indicating that no certificate can be issued. This process ensures that certificates are generated only for qualified participants, promoting fairness, accuracy, and integrity in the certification process while minimizing manual verification.

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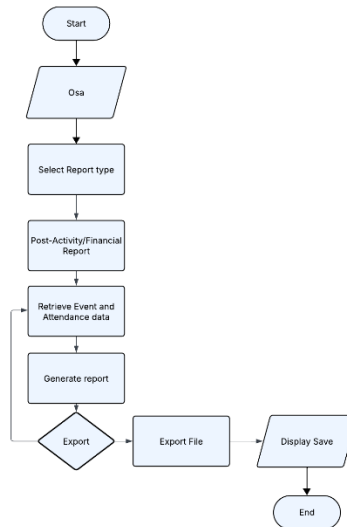


Figure 13: Procedural Workflow of Reports Module

Figure 13 presents the process of report generation within the system, primarily handled by the Office of Student Affairs (OSA). The process begins when the user selects the desired type of report, such as attendance reports or event summaries. The system then retrieves relevant data from the database, including event details and attendance records. After collecting the necessary information, the system processes and organizes the data into a structured report format. Once the report is generated, the user is given the option to export the report into a file for storage or sharing purposes. If the export option is selected, the system saves and displays the generated report accordingly. This process ensures that reports are systematically generated, easily accessible, and exportable, supporting efficient monitoring, evaluation, and documentation of events.

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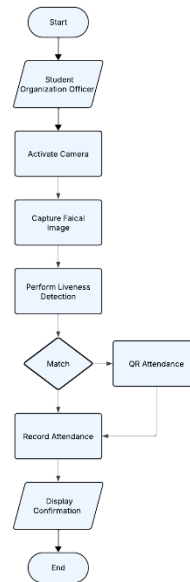


Figure 14: Procedural Workflow of Facial Recognition with Liveness detection Module

Figure 14 presents the process of recording attendance using facial recognition technology. The process begins by activating the system's camera to capture the user's facial image. The system then performs liveness detection to verify that the captured image is from a real person and not a spoofed or static image. After successful validation, the system compares the captured facial data with the stored facial records in the database. If a match is found, the system records the attendance and provides confirmation to the user. If no match is detected, the attendance is not recorded. This process enhances the security and reliability of attendance tracking by preventing fraudulent entries while ensuring accurate and automated verification of participants.

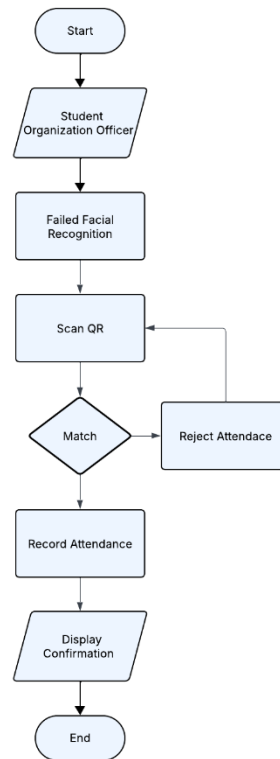


Figure 15: Procedural workflow of QR Module

Figure 15 presents the alternative method of recording attendance using QR codes. This process is typically used when facial recognition is unsuccessful or unavailable. The system prompts the user to scan a QR code associated with the event. Once scanned, the system verifies the validity of the QR code by checking it against stored event and user data. If the scanned information is valid, the system records the attendance and displays a confirmation message. If the data is invalid or does not match, the system rejects the attendance attempt. This alternative method ensures continuity in attendance recording by providing a reliable backup mechanism, thereby improving system flexibility and usability.

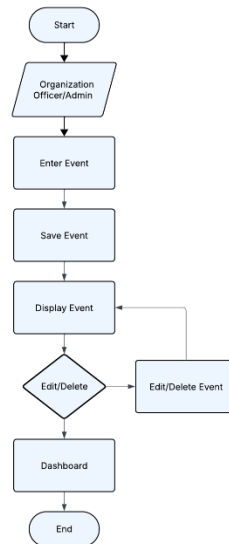


Figure 16: Procedural workflow of Event Module

Figure 16 explains how events are created and managed. The process starts with entering event details, which are then saved in the system. The event is displayed, allowing the user to review it. The user can choose to edit or delete the event if necessary. After completing the process, the system redirects to the dashboard. This ensures that event management is organized, editable, and user-controlled.

Data Modeling and Database Design

This section presents the data structure and database design of the proposed Web-Based Event Management System with Machine Learning–Based Facial Recognition Attendance. It includes data modeling techniques and diagrams, such as the Entity Relationship Diagram (ERD), to illustrate the relationships among system entities and how data is organized, stored, and managed. The section also explains how the database design supports system

functionalities, ensures data integrity, and enables efficient processing of user, event, attendance, and reporting data within the system.

Conceptual Data Mode

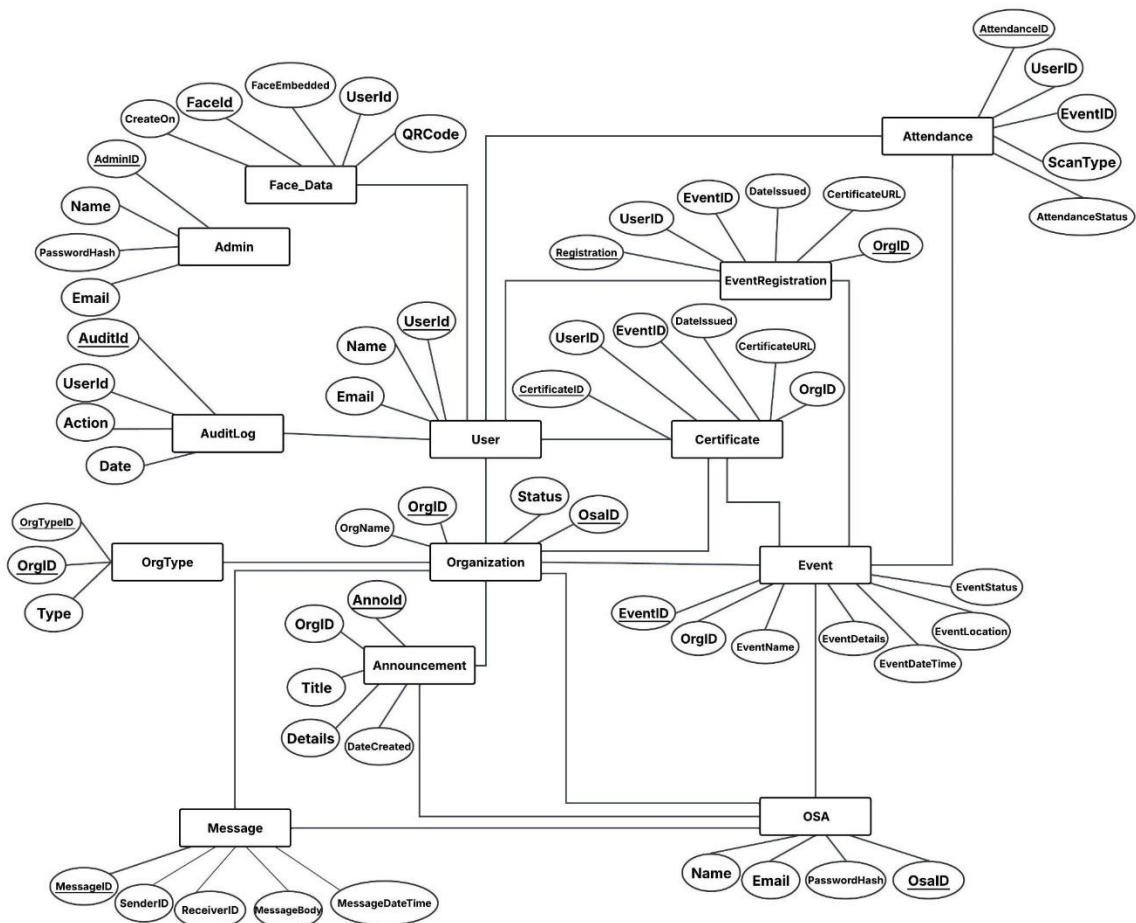


Figure 17: Conceptual Data Model

Figure 17 presents the Conceptual Data Model of the proposed Web-Based Event Management System with Machine Learning–Based Facial Attendance. The diagram illustrates the major entities involved in the system and the relationships among them at a high level, without including technical implementation details such as data types or constraints. The model identifies key entities including User,

Organization, Event, Event Registration, Attendance, Certificate, Announcement, Message, Face_Data, AuditLog, and OrgType. These entities represent the core components required to support the system's functionalities.

The User entity represents all system users, including students and authorized personnel, who interact with the system. Users are associated with Organizations, which are categorized through the OrgType entity, allowing the system to classify different types of organizations.

The Organization entity is responsible for managing events, which are represented by the Event entity. Users participate in events through the Event Registration entity, which records their enrollment or intent to attend. Actual participation is tracked through the Attendance entity, which stores records of attendance verified through biometric or QR-based methods.

The Certificate entity is linked to both users and events, representing the certificates issued to participants who meet the required attendance criteria. Communication within the system is handled through the Announcement and Message entities, which facilitate the dissemination of information and interaction among users.

Additionally, the Face_Data entity supports biometric authentication by storing facial recognition data, while the AuditLog entity records system activities for monitoring and security purposes. Overall, the conceptual data model provides a clear and simplified overview of how the major data components are interconnected, establishing the foundation for the system's database structure and supporting its key functionalities.

Entity Relationship Diagram

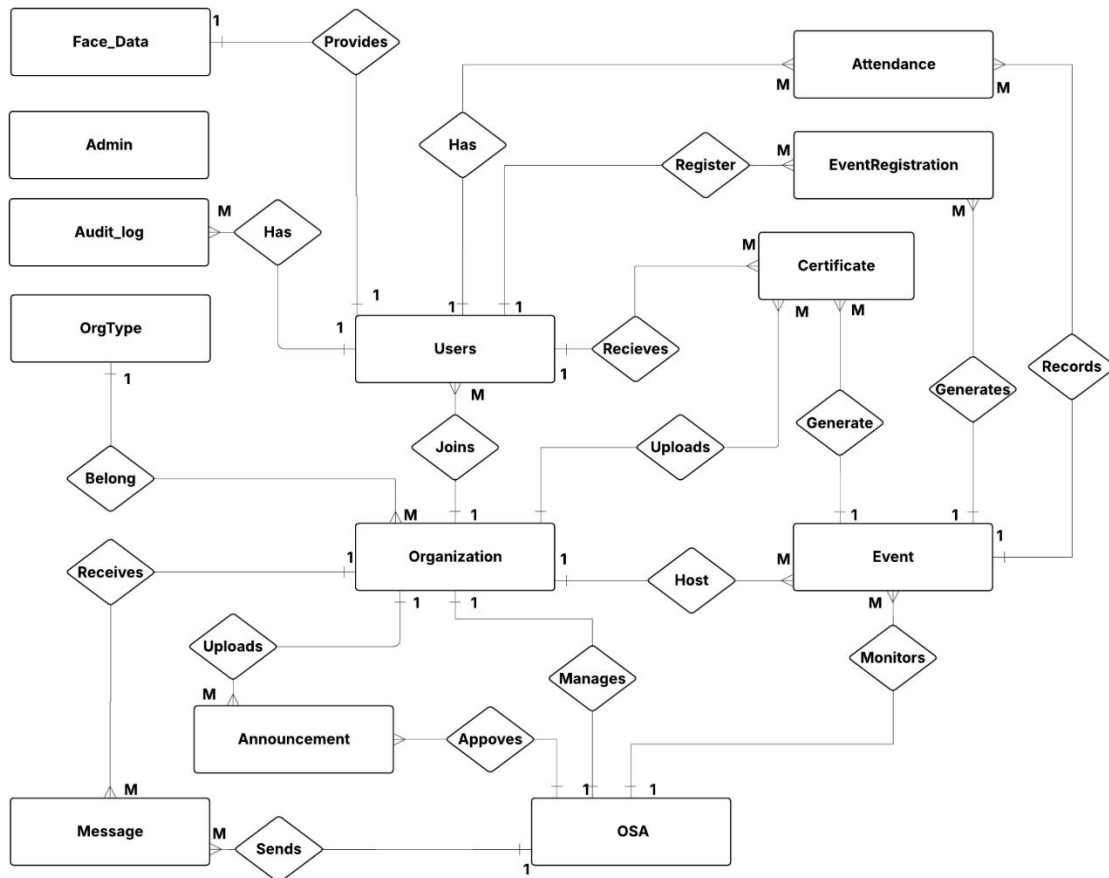


Figure 18: Entity Relationship Diagram

Figure 18 defines entity associations, relationship cardinalities, and data constraints to ensure data integrity, consistency, and efficient processing of event-related data within the system.

The User entity represents all system stakeholders, including students, student organization officers, Office of Student Affairs (OSA) personnel, and administrators. Each user is assigned specific roles that determine access privileges and system responsibilities. The User entity is associated with multiple

system components such as event participation, attendance tracking, certificate issuance, and communication, establishing several one-to-many relationships within the system.

The Organization entity represents student organizations registered in the system. Each organization is classified through the OrgType entity, forming a one-to-many relationship where one organization type can be assigned to multiple organizations. Organizations are responsible for managing events, announcements, and communication, establishing a one-to-many relationship between Organization and Event, as well as Organization and Announcement.

The Event entity represents activities created and managed by organizations. Each event contains relevant details such as event name, schedule, location, and status. The relationship between User and Event is modeled as a many-to-many relationship, as users can participate in multiple events and each event can have multiple participants. This relationship is resolved through the EventRegistration entity, which records user participation and registration details.

The Attendance entity stores records of user participation in events, including timestamps and attendance status. It establishes a one-to-many relationship between User and Attendance, and between Event and Attendance, ensuring that each attendance record is linked to a valid user and event. This entity supports attendance verification using biometric and QR-based methods.

The Certificate entity represents certificates issued to users who meet participation requirements. It is associated with both User and Event entities, forming one-to-many relationships where a user may receive multiple certificates from different events. This ensures that certificate issuance is properly linked to verified attendance records.

The Message and Announcement entities support communication within the system. The Announcement entity allows organizations or OSA to broadcast information to users, while the Message entity facilitates direct communication between system users. These entities establish one-to-many relationships with their respective senders and receivers.

The Face_Data entity is linked to the User entity and stores biometric data used for facial recognition. This relationship is typically one-to-one or one-to-many, depending on whether multiple facial records are stored per user. The AuditLog entity records system activities and user actions, supporting monitoring, security, and accountability within the system.

The ERD establishes relationship cardinalities and enforces data constraints through primary and foreign key relationships. These constraints ensure referential integrity by linking records across entities, preventing invalid or inconsistent data entries.

By defining structured entity relationships and enforcing data constraints, the ERD supports reliable data storage, accurate attendance tracking, efficient event management, and comprehensive reporting. Figure 18 illustrates the ERD, showing the logical relationships among system entities and ensuring proper data organization within the proposed system.

Logical Database Design

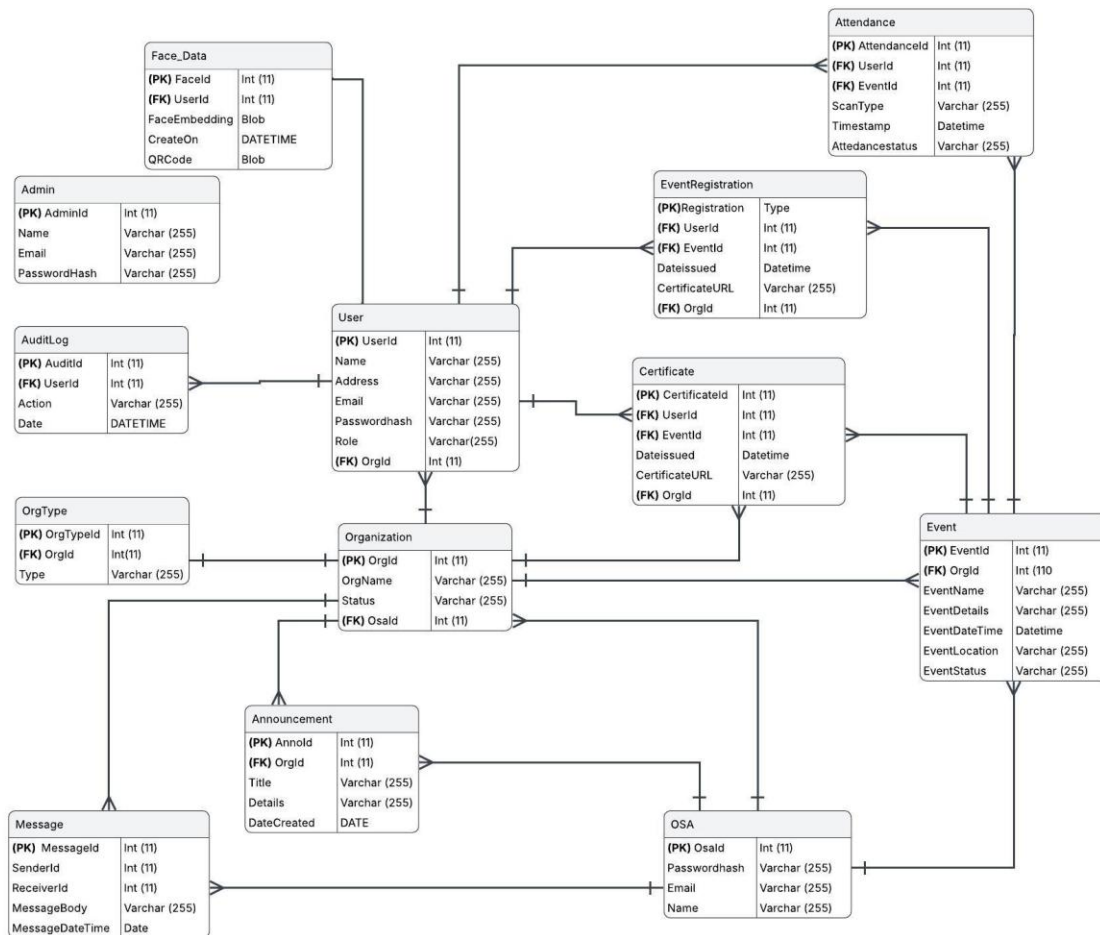


Figure 19: Logical Database Design

Figure 19 translates the Figure 18 Entity Relationship Diagram into a set of well-defined tables, attributes, primary keys, and foreign keys to support system implementation.

The User table serves as the central entity in the database, containing user information and linking to multiple related tables such as Organization, EventRegistration, Attendance, Certificate, and Message. This structure enables

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users to interact with various system functions, including joining organizations, participating in events, and receiving certificates.

The Organization table stores information about student organizations and is linked to the OrgType table, which classifies organizations into specific categories. Organizations are also connected to the Event table, where event details such as name, schedule, location, and status are stored. This establishes a one-to-many relationship where one organization can manage multiple events.

The EventRegistration table resolves the many-to-many relationship between users and events by recording registration details. The Attendance table captures actual participation data, including timestamps and attendance status, ensuring accurate tracking of user attendance through biometric and QR-based methods.

The Certificate table manages certificate issuance and is linked to both users and events, ensuring that certificates are generated only for qualified participants. Communication is supported through the Announcement and Message tables, which store broadcasted information and direct user interactions.

The Face_Data table stores biometric information used for facial recognition, linking it to the User table for secure identity verification. The AuditLog table records user activities and system events, providing support for monitoring, auditing, and security enforcement.

The logical database design establishes primary and foreign key relationships that enforce referential integrity, reduce data redundancy, and ensure consistency across all system tables. By organizing data into structured relations, the design supports efficient data retrieval, processing, and management.

Overall, the logical database design provides a clear and implementable structure that supports all system functionalities, ensuring reliable data handling and scalability for the proposed system.

System Security and Data Privacy Design

The Web-Based Event Management and Facial Recognition Attendance System implement security and data privacy measures to protect user information and ensure the system's safe operation. Since the system processes personal data, including student profiles, attendance records, and facial recognition data, safeguards are in place to prevent unauthorized access, misuse, and data loss. These measures support the responsible handling of information and align with institutional data protection requirements.

To secure access, the system uses authentication and role-based access control for three user groups: students, organization officers, and OSA administrators. Each user must log in with valid credentials, and access to features is limited by assigned roles. Organization officers can only manage and view records for their own organization, while OSA administrators have oversight access across all organizations. This reduces the risk of unauthorized viewing or modification of sensitive records. User activities in the admin panel are also recorded through audit logs to strengthen accountability and support monitoring of critical actions.

For data protection, the system secures information during transmission using encrypted communication protocols and applies secure database controls to protect stored records. Sensitive credentials, such as passwords, are not stored in plain text and are protected using secure hashing. In addition, regular data backup procedures help preserve records and support recovery in case of system failure. Overall, these security and privacy measures ensure that the system maintains

confidentiality, integrity, and availability of data while supporting reliable event management and attendance monitoring.

Access Control and Authentication Mechanisms

The system implements access control and authentication mechanisms to ensure that only authorized users can access specific system functions and data. Three primary user roles are supported within the platform: students, organization officers, and Office of Student Affairs (OSA) administrators. Each user type is assigned a unique account and corresponding access privileges based on their role and responsibilities within the system.

Authentication is enforced through a secure login process requiring registered credentials before access is granted. Once authenticated, users are directed to role-specific dashboards that limit their access to permitted features. Students can view event information, mark attendance, and download certificates. Organization officers can manage events, monitor attendance, and generate reports related only to their respective organizations. OSA administrators have broader access privileges, allowing them to oversee all organizations, manage system records, and monitor activities across the platform.

Role-based access control (RBAC) is implemented to restrict system actions and data visibility based on user permissions. This ensures that users can access only information relevant to their roles, preventing unauthorized viewing or modification of data. By enforcing structured authentication procedures and permission-based access levels, the system maintains data security, protects sensitive information, and ensures accountability for all system interactions.

Data Encryption and Protection Measures

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The Web-Based Event Management and Facial Recognition Attendance System implements data encryption and protection mechanisms to safeguard sensitive user information and ensure confidentiality during data transmission and storage. These measures are designed to prevent unauthorized access, protect personal and attendance data, and maintain the integrity of system records throughout all operations.

The system uses encrypted communication protocols to secure data exchanged between client devices and the server. This ensures that information such as login credentials, attendance submissions, and system requests cannot be intercepted, altered, or accessed by unauthorized parties during transmission. Encryption helps maintain secure communication channels and protects sensitive information while the system is in use.

To protect stored data, secure database management practices are implemented. Access to the database is restricted through authentication controls and permission settings, ensuring that only authorized administrators can view or modify records. Sensitive information is validated before storage, and database safeguards are applied to prevent unauthorized retrieval, modification, or deletion of data. These protections help maintain the reliability and integrity of attendance records, user information, and system-generated reports.

The system also includes data backup mechanisms to prevent data loss caused by technical failures or unexpected incidents. Regular backups support data recovery and ensure continuity of operations. By combining encrypted transmission, secure storage, controlled access, and backup procedures, the system maintains the confidentiality, integrity, and availability of information, ensuring that user data remains protected throughout system operation.

Data Privacy Act Compliance (RA 10173)

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The system complies with the provisions of the Data Privacy Act of 2012 (Republic Act No. 10173) by implementing technical, organizational, and physical safeguards to protect personal data processed within the platform. Since the system collects and processes sensitive information such as student identities and facial images, data privacy and protection were considered essential requirements throughout the system design and implementation phases.

To ensure compliance, the system applies data minimization principles by collecting only information necessary for event management, attendance verification, and reporting. User data, such as names, student identifiers, and facial templates are securely stored in the database with restricted access controls. Authentication mechanisms are enforced to ensure that only authorized users, such as administrators and organization officers, can access or modify records. Role-based access controls are implemented to limit system functions based on user responsibility levels.

Data transmission between client devices and the server is protected using encrypted communication protocols to prevent interception or unauthorized access. Additionally, passwords are securely stored using hashing to protect user authentication data. The system also maintains an audit logging feature in the administrative module to monitor user activities and track system interactions, supporting accountability and helping detect unauthorized actions.

Organizational safeguards are also observed by ensuring that users are informed about the purpose of data collection and how their information will be used. The system is designed to allow administrators to manage records responsibly, including updating or removing data when necessary in accordance with institutional policies. Furthermore, backup procedures are implemented to prevent data loss while maintaining data integrity and confidentiality.

By integrating these security and privacy measures, the system aligns with the core principles of the Data Privacy Act of 2012, including transparency, legitimate purpose, and proportionality. These safeguards ensure that personal data is handled responsibly, securely, and ethically, thereby protecting users' rights while supporting the system's operational objectives.

Audit Logging and Activity Monitoring

The system incorporates an audit logging and monitoring mechanism to ensure accountability, traceability, and security of system activities. An audit trail module is integrated into the administrative panel to record and track significant system events and user actions. This feature allows administrators to monitor system usage, detect irregular activities, and maintain a transparent record of operations performed within the platform.

The audit log automatically captures essential information for each recorded event, including user identification, the action performed, the affected module or record, the date and time stamp, and the system response status. Logged activities include administrative logins, event creation and modification, attendance validation, data updates, report generation, and other critical system operations. These logs are securely stored in the database to preserve historical records and support system monitoring.

This functionality enhances system reliability and security by enabling administrators to review activity histories, investigate discrepancies, and verify system integrity. In cases of errors, unauthorized access attempts, or data inconsistencies, the audit trail provides detailed evidence for troubleshooting and corrective action. Furthermore, maintaining a structured audit log supports compliance with institutional data management standards and promotes responsible use of the system.

Overall, the audit logging and monitoring component strengthens system governance by ensuring that all significant interactions within the platform are recorded, traceable, and reviewable, thereby supporting transparency, accountability, and operational control.

Implementation Environment and Technology Stack

The Web-Based Event Management and Facial Recognition Attendance System was developed using a combination of software tools, programming languages, and computing platforms to ensure reliable performance, accessibility, and efficient data processing. Development was conducted on a standard Windows computer, which served as the primary platform for coding, testing, and deployment. By implementing the system as a web-based application, cross-platform accessibility was achieved, eliminating the need for specialized client-side installation.

The frontend interface was developed using HTML, CSS, and JavaScript, enabling responsive layouts, interactive features, and intuitive navigation. These technologies provided a clean and user-friendly interface suitable for students, organization officers, and administrators. The backend was implemented in PHP, handling server-side processing, user authentication, request management, and communication among system components.

A MySQL database was used to store and organize system data, including user records, event information, attendance logs, facial images, and generated reports. MySQL was selected for its reliability, scalability, and compatibility with PHP-based applications. The development server environment was configured using XAMPP, which integrates the Apache web server, PHP interpreter, and MySQL database services into a unified platform for efficient development and testing.

Facial recognition functionality was implemented by integrating a machine learning library capable of processing captured images and matching them against stored facial templates. This module performs real-time identity verification and automatically records results in the attendance database. To ensure system reliability, a QR code-based fallback mechanism was incorporated to allow attendance recording when facial recognition cannot be performed due to environmental or hardware limitations.

System testing and deployment were conducted using modern web browsers such as Google Chrome and Microsoft Edge to verify compatibility, responsiveness, and stable performance across different devices. The system was designed to operate on commonly available hardware, including desktop computers, laptops, tablets, and smartphones equipped with cameras for image capture. Overall, the selected implementation environment and technology stack provide a stable, secure, and scalable foundation that supports efficient event management, accurate attendance tracking, and reliable data processing within the institution.

Software Requirements

The Event Management and Facial Recognition–Based Attendance System requires reliable software components to ensure efficient operation, secure data processing, and consistent user access. The system operates within a web-based environment hosted on an operating system such as Windows or Linux, which manages application execution and system services. A web server, such as Apache running on XAMPP or a similar platform, processes client requests and delivers system pages to users in real time.

For the frontend, the system uses HTML, CSS, and JavaScript to create an interactive, user-friendly environment. PHP is used as the backend programming

language to handle system logic, process requests, and manage communication between the interface and the database. Data is stored and organized using a MySQL database management system, which maintains essential records such as user accounts, event details, attendance logs, and facial recognition data. This ensures structured storage, efficient retrieval, and reliable data management.

The system integrates facial recognition functionality via compatible libraries or application programming interfaces that process and match captured facial images for identity verification. Users can access the platform using modern web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox, eliminating the need to install additional software on client devices.

To maintain system security and integrity, the software environment implements authentication controls, input validation, and encrypted communication protocols to protect sensitive information. Routine database backups and software updates are also implemented to maintain system stability and prevent data loss. Overall, these software requirements provide a stable and secure environment that supports accurate attendance monitoring, efficient event administration, and accessible system usage for administrators, organization officers, and students.

Hardware Requirements

The Event Management and Facial Recognition–Based Attendance System requires hardware that supports secure operations, handles multiple users simultaneously, and accurately verifies identities during events. Since most events are expected to have 100 or fewer participants, the hardware setup is designed to be practical, affordable, and efficient while still delivering reliable performance. The system needs a main computer or server to run the web application, manage communication between users and the system, and process attendance and facial

recognition tasks. A device with at least an Intel Core i3 processor (or its equivalent), 8 GB of RAM, and a 256 GB SSD is sufficient to support simultaneous access by students, organization officers, and administrators, as well as real-time image processing and report generation for small to medium-sized events.

Storage hardware is also necessary to keep attendance records, user information, event details, and captured facial images. Using SSD storage helps the system retrieve data quickly, while optional external drives or cloud storage can be used for backups to prevent data loss. Regular backups help ensure that institutional records remain safe and accessible. In addition, a stable network connection is required for devices to communicate smoothly with the system. An internet speed of around 5–10 Mbps, supported by a router or campus network, is generally enough to handle multiple users accessing the system simultaneously and maintain uninterrupted data transmission during attendance recording.

Client hardware devices, such as desktop computers, laptops, tablets, iPads, and smartphones, allow authorized users to access the system through standard web browsers without requiring specialized equipment. Recommended minimum client specifications include a dual-core processor, 4 GB RAM, and an updated browser. Additionally, camera-enabled devices with at least 720p resolution are required at event locations to capture facial data accurately for identity verification. With these adjusted specifications, the hardware environment remains scalable, reliable, and cost-efficient while still supporting accurate attendance monitoring, secure data processing, and smooth system operation for events with relatively small participant counts.

Development Tools and Programming Technologies

The development of the Web-Based Event Management and Facial Recognition Attendance System utilized a set of software tools and technologies

selected for their reliability, compatibility, and efficiency in building web-based applications. These tools supported the system's design, coding, testing, and deployment processes, ensuring that the platform functions smoothly and meets performance requirements.

For the frontend development, the system used HTML, CSS, and JavaScript to create responsive layouts, interactive elements, and a user-friendly interface. These technologies enabled the system to run on different devices and screen sizes while maintaining usability for students, organization officers, and administrators. Backend development was implemented using PHP, which handled server-side logic, user authentication, request processing, and communication between the interface and the database.

The system's data storage and management were handled using MySQL, a relational database management system chosen for its stability, scalability, and seamless integration with PHP. For local development and testing, XAMPP was used as the server environment because it provides an integrated package of Apache, MySQL, and PHP. This setup simplified configuration and allowed efficient testing before deployment.

To ensure compatibility and performance, testing was conducted using modern web browsers, including Google Chrome and Microsoft Edge. The system was designed to operate on commonly available hardware devices, including desktops, laptops, tablets, and smartphones equipped with cameras for facial recognition. Overall, the selected development tools and technologies provided a stable, flexible environment that enabled efficient system development, reliable operation, and future scalability.

Deployment Environment

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The Web-Based Event Management and Facial Recognition Attendance System is deployed within the institutional environment of the National Aviation Academy of the Philippines and operates via a web-based platform accessible via a local network or an internet connection. The deployment environment consists of a server computer that hosts the web application, database, and system services, allowing authorized users to access the system through standard web browsers without requiring additional software installation. This setup enables centralized system management while supporting simultaneous access by multiple users, including students, organization officers, and administrators.

The server environment runs on a supported operating system, such as Windows or Linux, and uses a web server to handle client requests, process transactions, and deliver system pages. The database is hosted on the same server or a dedicated database server to store user records, attendance logs, event information, and generated reports securely. Client devices connect to the system using commonly available hardware such as desktop computers, laptops, tablets, or smartphones with internet connectivity and camera capability for facial recognition attendance.

The deployment configuration is designed to support stable performance, secure data communication, and reliable accessibility across different devices and locations within the institution. By centralizing system services and data storage, the deployment environment ensures efficient system maintenance, consistent updates, and controlled access management, allowing administrators to monitor and manage system operations effectively.

Chapter IV

RESEARCH METHODOLOGY, RESULTS, AND DISCUSSION

This chapter presents the results and analysis of the development and evaluation of the Web-Based Event Management System with Machine Learning–Based Facial Attendance for Student Organizations of the National Aviation Academy of the Philippines. It focuses on examining the system’s performance, accuracy, and usability based on actual implementation and assessment activities conducted during selected student organization events.

The discussion covers findings from system testing, user evaluations, and performance measurements, including the effectiveness of the facial recognition attendance mechanism with a QR code fallback option. The chapter also examines the efficiency of event management processes, attendance recording, report generation, and certificate issuance compared to traditional manual methods. System quality is evaluated using ISO/IEC 25010:2023 software quality standards, ensuring a structured and objective assessment.

The results are interpreted in relation to the study’s objectives to determine whether the proposed system meets functional, technical, and user requirements. Through the presentation of empirical data and analytical discussion, this chapter establishes the reliability, effectiveness, and practical value of the system and provides a foundation for the conclusions and recommendations in the succeeding chapter.

Research Design

This study used a Design Science Research approach combined with a mixed-method evaluation to develop and assess a Web-Based Event Management System with Machine Learning Based Facial Attendance for student

organizations at the National Aviation Academy of the Philippines. This approach allowed the study to create a functional system while also generating insights into its reliability, efficiency, and user experience.

The research followed the core DSR stages: identifying problems, designing and developing the system, demonstrating it in real events, evaluating its performance, and sharing the results. Challenges in the current manual attendance and event management process included slow recording, data errors, difficulty tracking participation, and time-consuming report generation. The proposed system addresses these issues by integrating facial recognition for attendance, a QR code backup, and automated management features for scheduling, reporting, and certificate issuance.

The evaluation combined quantitative and qualitative methods. Quantitative data measured attendance accuracy, system response time, report generation speed, and operational efficiency. Qualitative data captured user experiences, challenges, and suggestions to understand usability and acceptance. A comparative assessment with the manual process highlighted measurable improvements in efficiency, accuracy, and user satisfaction.

Overall, the research design focused on developing a practical, user-centered system and testing it in real settings to determine whether automated attendance and event management could enhance the operations of student organizations while providing a seamless and reliable experience for all users.

Data Gathering Technique

Data for this study were collected using a combination of system-generated records, structured survey instruments, and qualitative feedback forms to ensure a comprehensive evaluation of the proposed Web-Based Event Management System with Machine Learning–Based Facial Attendance. Data gathering was

conducted during selected student organization events to capture real-world system performance and obtain authentic responses from actual users.

Quantitative data related to attendance accuracy and system efficiency were gathered directly from system logs and automated records. These included attendance timestamps, facial recognition success rates, QR code fallback usage, report generation time, and certificate issuance records. All data were automatically stored in the centralized database, ensuring accuracy and consistency for analysis.

To evaluate system quality and user experience, a structured survey questionnaire based on ISO/IEC 25010:2023 software quality standards was administered to students, organization officers, and administrators after system use. The survey measured functional suitability, performance efficiency, usability, reliability, security, compatibility, maintainability, flexibility, and safety using a Likert-scale format. Responses were collected electronically through the system interface.

Qualitative data were gathered through open-ended feedback forms embedded within the system. Participants were encouraged to share their experiences, concerns, and suggestions regarding the facial recognition process, QR code fallback, event management features, and overall usability. These responses were analyzed using thematic categorization to identify recurring issues, strengths, and recommendations for system improvement.

For comparative analysis, operational data from the traditional manual attendance process were collected, including average attendance recording time, frequency of data discrepancies, and administrative workload indicators. These were compared with system-generated data to assess improvements in efficiency, accuracy, and reliability.

All data collection procedures were conducted in accordance with ethical research standards. Participant identities were anonymized, and access to collected data was restricted through role-based authentication mechanisms to ensure confidentiality and data security.

Through the integration of automated system records, structured surveys, qualitative feedback, and comparative operational metrics, the study ensured that data gathering procedures were systematic, reliable, and aligned with the objectives of evaluating the effectiveness and usability of the proposed system.

Statistical Treatment of Data

The data gathered in this study were analyzed using appropriate descriptive and comparative statistical techniques to address the research objectives and support objective interpretation of results. Both quantitative and qualitative data analyses were conducted to evaluate the effectiveness, accuracy, and usability of the proposed system.

For quantitative data obtained from system evaluations and user surveys, descriptive statistics were applied to summarize participant responses. These included the computation of frequencies, percentages, means, and weighted means to determine overall system performance and user perception levels. The weighted mean was used to analyze responses collected through Likert-scale instruments, particularly for assessing system quality based on ISO/IEC 25010:2023 software quality characteristics, including functional suitability, performance efficiency, usability, reliability, security, compatibility, maintainability, flexibility, and safety.

To evaluate system effectiveness, comparative statistical analysis was employed to compare operational performance between the proposed system and the traditional manual attendance method. Metrics such as average attendance

recording time, frequency of data discrepancies, report generation speed, and administrative workload indicators were analyzed using mean differences and percentage improvement calculations to identify measurable improvements brought by the system.

Qualitative data gathered from open-ended feedback forms were analyzed using thematic analysis and frequency-based categorization. Responses were reviewed, grouped into recurring themes, and interpreted to identify common user experiences, issues, and recommendations related to facial recognition accuracy, QR code fallback usage, and overall system usability.

To ensure the reliability of the survey instruments, internal consistency testing, such as Cronbach's Alpha, was applied to the ISO-based system quality evaluation questionnaire where applicable. This helped verify that the survey items consistently measured the intended system quality attributes.

All statistical analyses were performed using appropriate statistical tools to ensure accuracy, consistency, and reproducibility of results. The applied statistical treatments enabled an objective assessment of system performance, user satisfaction, operational efficiency, and overall effectiveness of the proposed Web-Based Event Management System.

Statistical Formulas Used

1. Frequency (f)

Frequency refers to the number of times a specific response occurs within a dataset. It was used to determine how many respondents selected particular rating options in the system evaluation surveys.

f = Number of times a response occurs

2. Percentage (%)

Percentage represents the proportion of respondents selecting a specific response relative to the total number of respondents. It was used to express frequency data in a standardized and easily interpretable form.

$$\% = \left(\frac{f}{N} \right) \times 100$$

f = frequency of a response

N = total number of respondents

Percentage was used to interpret attendance completion rates, levels of user satisfaction, and the distribution of responses across survey categories. Converting frequency values into percentages allowed clearer comparison among respondent groups and improved the presentation of findings in a form that is easily understood by stakeholders and decision-makers.

3. Mean (Arithmetic Mean)

The arithmetic mean refers to the average value obtained by adding all numerical responses and dividing the total by the number of observations. In this study, the mean was used to compute average ratings related to system performance, attendance efficiency, and overall user experience. Since Likert-scale responses were treated as interval data, the mean served as an effective measure for comparing perceptions across different user groups.

$$\bar{X} = \frac{\sum X}{N}$$

Where:

$\bar{X}$ = mean

$\sum X$ = sum of all scores

N = number of respondents

The mean was used to compute the average system performance ratings and overall user evaluation scores for the Web-Based Event Management System with Machine Learning–Based Facial Attendance. Since the evaluation responses were collected using Likert-scale instruments and treated as interval-level data, the mean served as a reliable measure of central tendency for comparing evaluation results among students, organization officers, and administrators.

4. Weighted Mean (For Likert Scale)

The weighted mean is a statistical measure used to determine the overall level of responses by considering how often each rating appears. This method assigns appropriate weight to each Likert-scale value based on its frequency, allowing a more accurate interpretation of scaled responses.

$$WM = \frac{\sum(f \times x)}{N}$$

Where:

f = frequency of each rating

x = Likert scale value

N = total responses

In this study, the weighted mean was used to analyze Likert-scale responses obtained from the ISO/IEC 25010:2023 system quality evaluation and user satisfaction surveys. Since respondents rated system features using a scaled format, the weighted mean provided a precise summary of overall system performance, usability, reliability, and security levels.

5. Standard Deviation (SD)

Standard deviation is a measure of variability that describes how far individual responses deviate from the mean. It indicates whether responses are closely grouped or widely dispersed.

$$SD = \sqrt{\frac{\sum(X - \bar{X})^2}{N}}$$

Standard deviation was applied to assess the consistency of responses provided by students, student organization officers, and administrators. A smaller standard deviation indicates agreement among respondents, while a larger value suggests varied perceptions. This measure was useful in determining the reliability of user evaluations of the proposed system.

6. Percentage Improvement (For System Comparison)

Percentage improvement measures the relative change between two values and expresses the difference as a percentage. It is used to determine whether a process has improved or declined after system implementation.

$$\text{Improvement} = \left(\frac{\text{New} - \text{Old}}{\text{Old}} \right) \times 100$$

This measure was utilized to compare operational indicators between the proposed web-based system and the traditional manual attendance method. Indicators such as attendance recording time, report generation speed, and administrative workload were analyzed to quantify system improvements.

7. Percentage Improvement (For System Comparison)

Cronbach's Alpha is a statistical measure used to determine the internal consistency of a survey instrument. It evaluates how closely related a group of survey items are in measuring the same construct.

$$\alpha = \frac{k}{k-1} \left(1 - \frac{\sum \sigma_i^2}{\sigma_t^2} \right)$$

Where:

- k = number of items
- σ_i^2 = variance of each item
- σ_t^2 = total variance

In this study, Cronbach's Alpha was applied to test the reliability of the ISO-based system quality evaluation questionnaire and user perception surveys.

Establishing acceptable reliability values ensured that the instruments consistently measured the intended system quality characteristics.

8. Cronbach's Alpha (Reliability Test)

Qualitative response analysis was conducted by examining the repetition of specific terms and expressions within narrative feedback provided by system users. This method focuses on determining which ideas appear most frequently across open-ended responses.

Frequency of word=Number of occurrences of keyword

In this study, Cronbach's Alpha was used to validate the reliability of user evaluation instruments completed by students and organization officers who utilized the event management and facial attendance features. Ensuring reliable measurement strengthened the credibility of findings related to system usability, accuracy of attendance recording, efficiency of event monitoring, and overall user satisfaction.

Population and Sampling Technique

The study utilized a purposive sampling approach in which participants were intentionally selected based on their direct involvement in the use and operation of the proposed Web-Based Event Management System with machine learning-based facial attendance at the National Aviation Academy of the Philippines (NAAP). This method was appropriate because the evaluation of the system required respondents who had actual experience using its event management, attendance monitoring, and reporting features.

The respondents consisted of students who participated in organization events, student organization officers responsible for managing events, and authorized administrators who accessed attendance records and system-

generated reports. These groups represent the primary users of the system and were therefore essential in determining whether the system achieved its objectives related to attendance accuracy, operational efficiency, and transparency.

Purposive sampling was applied because the study focused on assessing the system within its intended operational setting rather than generalizing results to other institutions. Selecting participants who directly interacted with the system ensured that the data collected were relevant, experience-based, and aligned with the research objectives.

Inclusion criteria required that respondents had authorized system access, participated in at least one event during the implementation period, and consented to participate in system evaluation activities. Individuals who did not actively use the system or had incomplete interaction records were excluded to maintain data reliability.

Result and Discussion

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS o

Summary

Conclusions

Recommendation

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Survey Questionnaire

Data Privacy and Confidentiality Statement

This survey is conducted as part of an academic research study titled:

“Web-Based Event Management System with Machine Learning-Based Facial Attendance for Student Organization of the National Aviation Academy of the Philippines.”

Participation in this study is voluntary. By proceeding with this questionnaire, you confirm that you have read and understood the information provided and agree to participate under the conditions described below.

All information collected in this survey will be used solely for academic research and system evaluation purposes. The data gathered will include your responses to system evaluation items and general demographic information such as your role (e.g., student organization member, officer, or adviser). No personally identifiable information (such as full name, student number, or contact details) will be required unless explicitly stated.

All responses will be treated with strict confidentiality and will be anonymized prior to analysis. Data will be stored securely and accessed only by the authorized researchers and academic advisers involved in this study.

This study complies with the **Data Privacy Act of 2012 (Republic Act No. 10173)** and follows institutional research ethics guidelines.

Participants may decline to answer any question or withdraw from the study at any time before submission without penalty.

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If you have concerns regarding your data privacy rights or the handling of your responses, you may contact the research proponent or the institution's designated data protection officer.

Consent Declaration

- ☐ I have read and understood Data Privacy and Confidentiality Statement.
- ☐ I voluntarily agree to participate in this research study.

Name (Optional): _____

SECTION A: Respondent Profile

1. Role:

- ☐ Student
- ☐ Student Organization Officer
- ☐ OSA
- ☐ Admin

Number of times you used the system this term:

- ☐ 1-2
- ☐ 3-5
- ☐ More than 5

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Rating Scale

Scale	Description
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

SECTION B - SYSTEM QUALITY EVALUATION

To evaluate the system based on ISO/IEC 25010:2023 software quality characteristics.

A. Functional Suitability

Criteria Definition:

The degree to which the system provides functions that meet stated and implied needs.

No.	Item	1	2	3	4
1	The system allows easy event creation and management.	☐	☐	☐	☐

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2	The facial attendance and QR feature works accurately.	☐	☐	☐	☐
3	The system records attendance correctly.	☐	☐	☐	☐
4	Reports generated by the system are accurate.	☐	☐	☐	☐

B. Reliability

Criteria Definition:

The degree to which the system performs specified functions under stated conditions for a specified period.

No.	Item	1	2	3	4
5	The system operates without crashes or errors..	☐	☐	☐	☐
6	Attendance data is saved reliably.	☐	☐	☐	☐
7	The system performs well during large events.	☐	☐	☐	☐

C. Usability

Criteria Definition:

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The degree to which the system can be used effectively, efficiently, and satisfactorily by users.

No.	Item	1	2	3	4
8	The interface is user-friendly.	☐	☐	☐	☐
9	Navigation within the system is easy.	☐	☐	☐	☐
10	The system is easy to learn and use.	☐	☐	☐	☐

D. Performance Efficiency

Criteria Definition:

The degree to which the system performs relative to the amount of resources used.

No.	Item	1	2	3	4
11	The system loads quickly.	☐	☐	☐	☐
12	Facial recognition processes attendance quickly.	☐	☐	☐	☐

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E. Security

Criteria Definition:

The degree to which the system protects information and data.

No.	Item	1	2	3	4
13	I feel my data is secure in the system.	☐	☐	☐	☐
14	The system protects attendance records from unauthorized access.	☐	☐	☐	☐

F. Compatibility/Portability

Criteria Definition:

The degree to which the system operates across different devices and environments.

No.	Item	1	2	3	4
15	The system works well on mobile devices.	☐	☐	☐	☐
16	The system works across different browsers.	☐	☐	☐	☐

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SECTION C – MACHINE LEARNING-BASED FACIAL ATTENDANCE

To assess the effectiveness of the facial recognition feature

No.	Criteria	1	2	3	4
1	Facial recognition accurately identifies users.	☐	☐	☐	☐
2	The system reduces proxy attendance.	☐	☐	☐	☐
3	Attendance recording is faster than manual methods.	☐	☐	☐	☐
4	The system performs well under different lighting conditions.	☐	☐	☐	☐
5	The system works even with minor facial changes (e.g., glasses).	☐	☐	☐	☐

SECTION D – EVENT MANAGEMENT EFFECTIVENESS

To evaluate the event management features

No.	Criteria	1	2	3	4
1	The system simplifies event organization.	☐	☐	☐	☐
2	The system improves attendance monitoring.	☐	☐	☐	☐

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3	The system helps in generating event reports efficiently.	☐	☐	☐	☐
4	The system enhances overall event management.	☐	☐	☐	☐

SECTION E – TECHNOLOGY ACCEPTANCE & IMPACT

To evaluate user acceptance and institutional impact of the system.

No.	Criteria	1	2	3	4
1	The system improves efficiency in attendance tracking.	☐	☐	☐	☐
2	I find the system useful in my role.	☐	☐	☐	☐
3	I am willing to use this system in future events.	☐	☐	☐	☐
4	The system enhances transparency in attendance.	☐	☐	☐	☐
5	Overall, I am satisfied with the system.	☐	☐	☐	☐

SECTION F- OPEN-ENDED QUESTIONS

To collect qualitative insights for system improvement.

1. What features of the system do you find most useful?
2. What improvements would you recommend?

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3. Do you have any concerns about facial recognition attendance?
4. How does the system compare to your previous method of attendance tracking?
5. What challenges did you encounter while using the system?
6. In what ways can the facial recognition feature be improved?
7. Do you think the system is reliable during large events? Why or why not?
8. How can the system better support student organizations in managing events?
9. What additional features would you like to see in future versions of the system?
10. How does the system impact the efficiency of event management in your organization?

Signature

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